Article

Modelling human blastocysts by reprogramming fibroblasts into iBlastoids

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Xiaodong Liu[1,2,3,11], Jia Ping Tan[1,2,3,11], Jan Schröder[1,2,3], Asma Aberkane[3], John F. Ouyang[4], Monika Mohenska[1,2,3], Sue Mei Lim[1,2,3], Yu B. Y. Sun[1,2,3], Joseph Chen[1,2,3], Guizhi Sun[1,2,3], Yichen Zhou[1,2,3], Daniel Poppe[5,6], Ryan Lister[5,6], Amander T. Clark[7,8,9,10], Owen J. L. Rackham[4], Jennifer Zenker[3] & Jose M. Polo[1,2,3] ✉

Human pluripotent and trophoblast stem cells have been essential alternatives to blastocysts for understanding early human development[1–4]. However, these simple culture systems lack the complexity to adequately model the spatiotemporal cellular and molecular dynamics that occur during early embryonic development. Here we describe the reprogramming of fibroblasts into in vitro three-dimensional models of the human blastocyst, termed iBlastoids. Characterization of iBlastoids shows that they model the overall architecture of blastocysts, presenting an inner cell mass-like structure, with epiblast- and primitive endoderm-like cells, a blastocoel-like cavity and a trophectoderm-like outer layer of cells. Single-cell transcriptomics further confirmed the presence of epiblast-, primitive endoderm-, and trophectoderm-like cells. Moreover, iBlastoids can give rise to pluripotent and trophoblast stem cells and are capable of modelling, in vitro, several aspects of the early stage of implantation. In summary, we have developed a scalable and tractable system to model human blastocyst biology; we envision that this will facilitate the study of early human development and the effects of gene mutations and toxins during early embryogenesis, as well as aiding in the development of new therapies associated with in vitro fertilization.

Mammalian embryogenesis begins with a totipotent zygote that can develop into a morula, followed by the formation of a blastocyst. As the embryo implants, cells of the epiblast (EPI) lineage within the blastocyst develop into the embryo proper and the amnion, whereas cells of the trophectoderm (TE) and primitive endoderm (PE) will eventually give rise to the placenta and yolk sac, respectively[1]. When cultured in vitro, EPI cells give rise to human embryonic stem (hES) cells[2]. Alternatively, adult cells can be reprogrammed into human induced pluripotent stem (hiPS) cells by transcription factor-mediated reprogramming[3]. These pluripotent cells can be differentiated into all cell types of the body and have been used to develop in vitro models to study early human development[4]. Although these simple stem cell models have provided unprecedented insights into the genetic and epigenetic requirements for human pluripotency, they provide limited information on the complex spatiotemporal molecular changes that occur as the blastocyst and peri-implantation human embryo develop. At present, the only way to study this is to incorporate information from blastocysts that have been donated to research following in vitro fertilization. The use of human blastocysts for research purposes poses a myriad of ethical, policy and technical constraints that, collectively, have limited our understanding of early human embryonic development. Recently, in vitro integrated models of mouse blastocysts, called blastoids, have been produced using two different approaches: by assembling pre-established stem cell lines[5–9], or by differentiating extended (or expanded) pluripotent stem (EPS) cells into blastocyst-like structures[10]. Despite mouse blastoids not having full developmental potential, they can model in vitro several aspects of mouse early development[7–10]. However, the generation of human blastoids has not been previously reported, to our knowledge. While studying human reprogramming intermediates, we unexpectedly discovered that those intermediates, when aggregated, could form cellular structures that have undergone cavitation. To determine the true nature of these structures, we performed a battery of molecular and functional assays. These assays demonstrated that the structures were blastocyst-like structures that could model several aspects of blastocyst biology. Therefore, as the structures were generated by reprogramming fibroblasts, we named them induced blastoids or 'iBlastoids'. To facilitate understanding of this paper, we use the term iBlastoids retrospectively when referring to these structures.

[1]Department of Anatomy and Developmental Biology, Monash University, Clayton, Victoria, Australia. [2]Development and Stem Cells Program, Monash Biomedicine Discovery Institute, Clayton, Victoria, Australia. [3]Australian Regenerative Medicine Institute, Monash University, Clayton, Victoria, Australia. [4]Program in Cardiovascular and Metabolic Disorders, Duke-National University of Singapore Medical School, Singapore, Singapore. [5]Australian Research Council Centre of Excellence in Plant Energy Biology, School of Molecular Sciences, The University of Western Australia, Perth, Western Australia, Australia. [6]The Harry Perkins Institute of Medical Research, Perth, Western Australia, Australia. [7]Department of Molecular Cell and Developmental Biology, University of California Los Angeles, Los Angeles, CA, USA. [8]Molecular Biology Institute, University of California Los Angeles, Los Angeles, CA, USA. [9]Jonsson Comprehensive Cancer Center, University of California Los Angeles, Los Angeles, CA, USA. [10]Eli and Edythe Broad Center of Regenerative Medicine and Stem Cell Research, University of California Los Angeles, Los Angeles, CA, USA. [11]These authors contributed equally: Xiaodong Liu, Jia Ping Tan. ✉e-mail: jose.polo@monash.edu

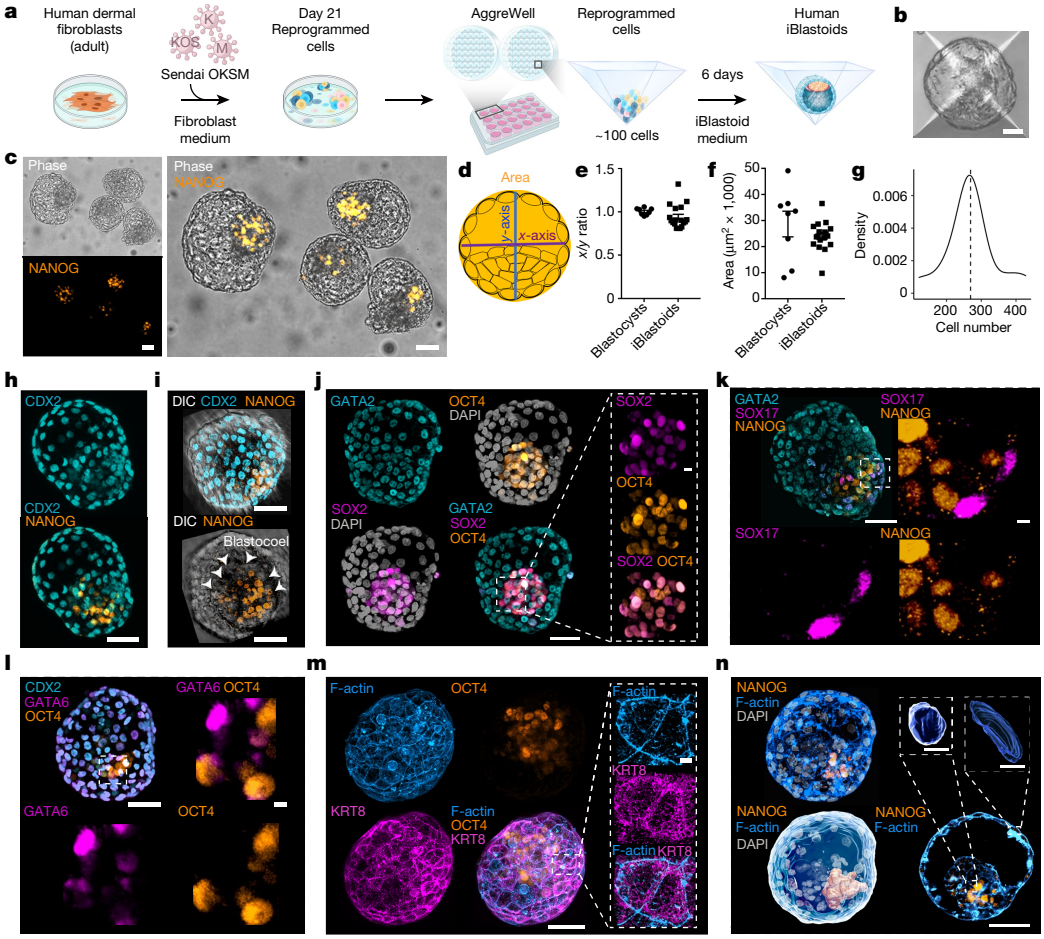


Fig. 1 | Generation of human iBlastoids. a, Schematic representation of the experiments. Sendai OKSM, Sendai viral vectors carrying human *OCT4*, *SOX2*, *KLF4*, and *c-MYC*. **b**, **c**, Representative phase-contrast images of an iBlastoid (**b**) and immunostaining for NANOG (**c**), $n = 5$. **d**–**f**, Measurement of *x*/*y* ratio (**d**, **e**) and area (**d**, **f**) of iBlastoids compared to blastocysts (data for blastocysts from refs. [12–19]); mean ± s.e.m., $n = 18$ (iBlastoids), $n = 8$ (blastocysts[12–19]). **g**, Distribution of total iBlastoid cell numbers (dashed line, mean); $n = 14$. **h**, **i**, Co-immunostaining of iBlastoids for CDX2 and NANOG (**h**) with DIC (**i**); arrowheads, blastocoel-like cavity; $n = 5$. **j**, Co-immunostaining of iBlastoids for GATA2, OCT4 and SOX2 (dashed area expanded to right); $n = 3$. **k**, Co-immunostaining of iBlastoids for GATA2, NANOG and SOX17 (dashed box expanded in other panels); $n = 2$. **l**, Co-immunostaining of iBlastoids for CDX2, OCT4 and GATA6 (dashed box expanded in other panels); $n = 2$. **m**, Co-immunostaining of iBlastoids for F-actin, OCT4 and KRT8 (dashed area expanded to right); $n = 2$. **n**, Co-immunostaining of iBlastoids for F-actin and NANOG (expanded views at top right show EPI-like (left) and TE-like cells (right), highlighting their morphological differences). Bottom left, 3D segmentation, $n = 2$. Scale bars, 50 μm (main images), 5 μm (expanded views).

Generation of human iBlastoids

We recently uncovered the transcriptional roadmap for reprogramming of human dermal fibroblasts (HDFs) into hiPS cells[11]. Notably, when reprogramming is performed entirely in fibroblast medium, cells with EPI, TE, and PE transcriptional signatures emerge by day 21 (Extended Data Fig. 1a). Immunostaining analysis with CDX2 (TE marker), OCT4 (EPI marker) and GATA6 (PE marker) confirmed that the 2D culture system contained a heterogeneous cell population (Extended Data Fig. 1b–e). This heterogeneity provided the opportunity to study the molecular interactions of early developmental signatures at scale. To acquire these cells, we initiated reprogramming with *OCT4* (also known as *POU5F1*), *KLF4*, *SOX2* and *c-MYC* (also known as *MYC*) as previously described[11], and transferred day 21 reprogrammed cells into the AggreWell system at a limiting dilution to allow 3D cellular interactions in a medium that could sustain the early developmental signatures. This led to the formation of 3D cellular structures, with some of them (iBlastoids) presenting blastocyst-like cavitation at day 6 (Fig. 1a, b). Immunofluorescence staining for NANOG showed positive staining in the cells adjacent to the cavity, which were surrounded by an outer layer of NANOG-negative cells (Fig. 1c). The diameters and areas of the iBlastoids were comparable to previously published measurements of human blastocysts at embryonic days 5–7 (E5–7) after fertilization[12–19] (Extended Data Fig. 1f, Fig. 1d–f) and the number of cells ranged from 100 to 400 (mean, 269), which is similar to that (about 240 cells) previously reported for human blastocysts at E5–7[19–22] (Fig. 1g). The efficiency with which iBlastoids formed ranged from 5.8% to 18% across three donors (Extended Data Fig. 1g, h). We observed cells that are not part of the structures attached to the microwell, and determined that these cells were refractory to reprogramming (Extended Data Fig. 1i); when cultured in fibroblast medium, they expressed the refractory cell markers *TWIST1* and *ZEB1* but not the pluripotency markers *NANOG* and *SOX2* (Extended Data Fig. 1j, k).

Spatial cellular localization of iBlastoids

Next, we performed co-immunostaining for the EPI marker NANOG and the TE marker CDX2[12,20], which revealed that the NANOG-positive cells were located exclusively in an inner cell mass (ICM)-like compartment, whereas CDX2-positive cells were found in the outer layer, which resembles the TE (Fig. 1h). The 3D reconstruction and overlay of

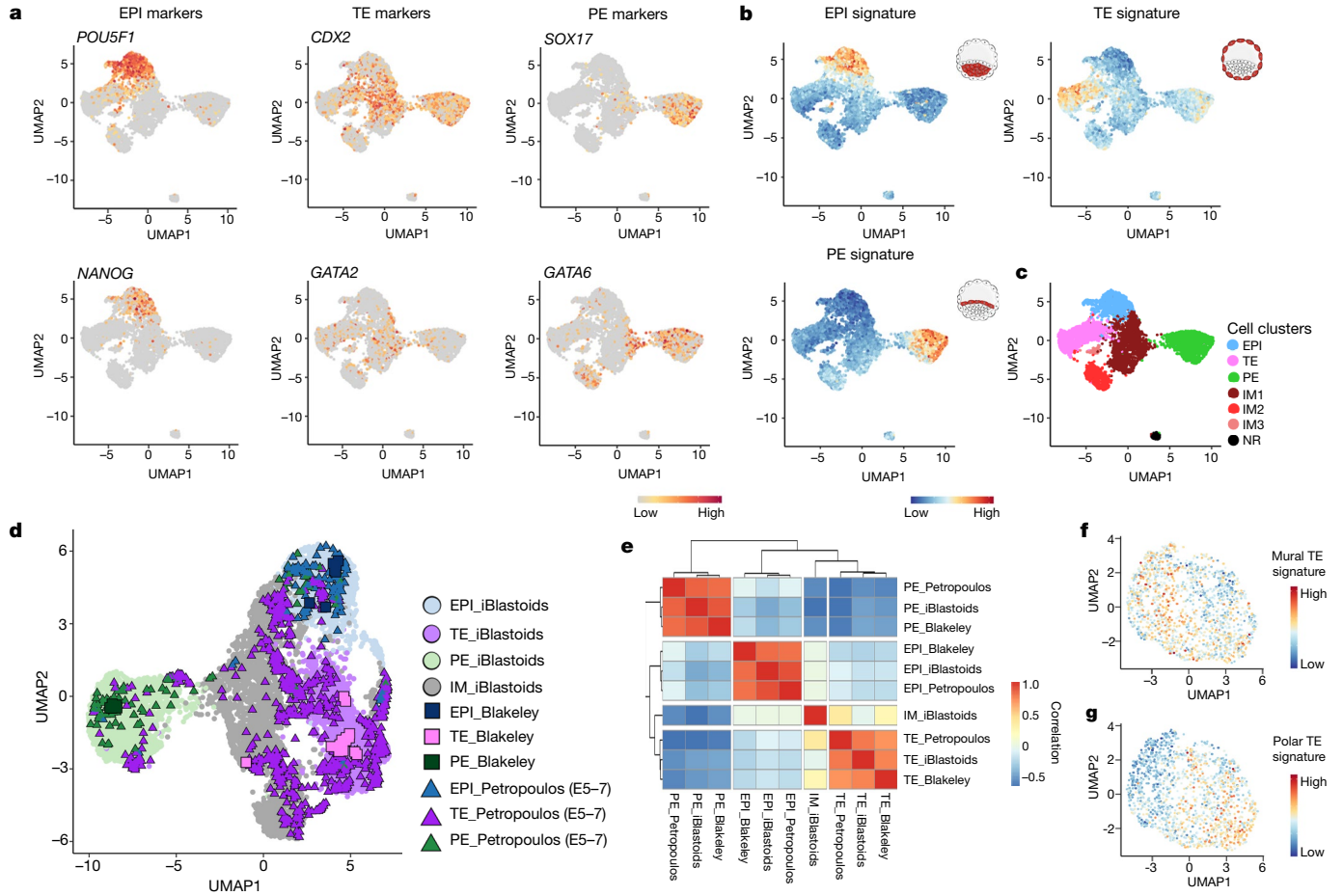


Fig. 2 | Single-cell transcriptomic profiling of iBlastoids. a, Expression of EPI markers (left; *POU5F1* and *NANOG*), TE markers (middle; *CDX2* and *GATA2*), and PE markers (right; *SOX17* and *GATA6*) for 6,858 cells from iBlastoid scRNA-seq library. **b**, Per-cell expression score for EPI, TE, and PE signatures on UMAP of iBlastoid scRNA-seq dataset. **c**, Unsupervised clustering of iBlastoid dataset with assigned cluster names. **d**, UMAP projection of integrated datasets (7,861 cells) showing iBlastoid EPI, TE, and PE-like cells (and clusters IM1–3, combined as IM_iBlastoids) together with EPI, TE, and PE cells from blastocysts from two published studies (Blakeley et al.[12] and Petropoulos et al.[13]). **e**, Pearson correlation analysis of iBlastoid EPI, TE, and PE-like clusters with annotated EPI, TE, and PE clusters from blastocysts as in **d**[12,13]. **f**, **g**, Per-cell expression scores of defined mural (**f**) and polar (**g**) TE signatures on iBlastoid scRNA-seq TE cluster.

differential interference contrast (DIC) images indicated the existence of a blastocoel-like cavity, similar to blastocysts[14,15,21] (Fig. 1i, Supplementary Videos 1, 2). Notably, similar results were obtained with fibroblasts from two additional donors (Extended Data Fig. 2a). As described for human blastocysts[19], the outer TE-like cells of iBlastoids were positive for the TE marker GATA2, whereas the EPI markers OCT4 and SOX2 were co-localized in the ICM-like compartments (Fig. 1j). Within the ICM-like compartment, we identified PE-like cells, marked by SOX17, at the periphery of the NANOG-positive cells (Fig. 1k, Extended Data Fig. 2b), similar to E6–7 blastocysts[15]. Subsequent analysis showed that the proportions of EPI-, PE-, and TE-like cells in the iBlastoids were similar to that found in blastocysts[20] (Extended Data Fig. 2c). Additional immunostaining revealed a 'salt and pepper' pattern of GATA6 (PE marker) and CDX2 (TE marker) in the TE-like compartment (Fig. 1l, Extended Data Fig. 2d). GATA6 has previously been ubiquitously detected in TE cells of blastocysts[21,22]. Notably, closer examination identified GATA6-positive cells (with weak CDX2 staining) neighbouring OCT4-positive cells in the ICM-like compartment, which suggests the presence of GATA6-positive PE-like cells (Fig. 1l, Extended Data Fig. 2d). As seen in blastocysts[21], we observed GATA3 staining in the TE-like cells (Extended Data Fig. 2e) and a dense keratin 8 (KRT8) filament network within the outer TE-like cells of the iBlastoids (Fig. 1m). Moreover, iBlastoids were positive for Ki67, a marker for cell proliferation (Extended Data Fig. 2f).

We then used F-actin (marked by fluorescence-labelled phalloidin) to label membrane contours and visualize cellular morphology. As in blastocysts[23], the compact NANOG-positive EPI-like cells had a more rounded columnar appearance, whereas the TE-like cells were flattened, highlighting that this model can recapitulate differences in the cellular architectures of EPI and TE cells (Fig. 1n, Extended Data Fig. 3c). Finally, a day-by-day time-course immunostaining analysis with GATA6, CDX2, and OCT4 suggested a combination of cell integration and cell-fate specification during the generation of iBlastoids (Extended Data Fig. 3a, b).

Single-cell transcriptomics of iBlastoids

To further characterize the transcriptional make-up of the iBlastoid's cells beyond just a few key genes, we performed single-cell RNA sequencing (scRNA-seq) on about 200 iBlastoids from two donors (Extended Data Fig. 4a, b, Supplementary Table 1; see Methods). Uniform manifold approximation and projection (UMAP) analysis indicated an even distribution without clustering on the basis of donor or cell cycle (Extended Data Fig. 4c, d). Although exogenous *KLF4* transcripts could be detected, there was minimal or no detection of the other reprogramming factors (Extended Data Fig. 4e). We then identified the putative EPI, TE, and PE cells using marker genes such as *NANOG* and *OCT4* for EPI; *CDX2*, *GATA2*, *GATA3* and *KRT8* for TE; and *SOX17* and

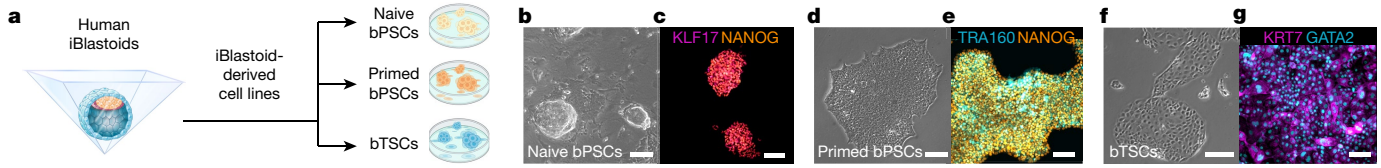


Fig. 3 | Derivation of stem cells from iBlastoids. a, Experimental design for derivation of naive bPSCs, primed bPSCs and bTSCs from iBlastoids. **b**–**g**, Phase-contrast and immunostaining analysis of naive bPSCs (**b**, **c**), primed bPSCs (**d**, **e**) and bTSCs (**f**, **g**). $n = 2$. Scale bars, 100 µm.

GATA6 for PE. *OCT4*- and *NANOG*-expressing cells occupied a different region in the UMAP space than the cells expressing *SOX17* and *GATA6* (Fig. 2a, Extended Data Fig. 4f). On this note, more cells expressed endogenous *OCT4* than *NANOG*, as observed in human blastocysts[13] (Extended Data Fig. 4g). However, the expression of *CDX2* and *GATA2* was more heterogeneous, which is similarly observed in E5–7 human blastocysts[13] (Fig. 2a, Extended Data Fig. 4g).

We further confirmed the presence of EPI-, TE-, and PE-like lineages in the iBlastoids, using gene signatures defined for each lineage in E5–7 human blastocysts[13] (Fig. 2b, Extended Data Fig. 5a, Supplementary Table 2). Furthermore, E5, E6, and E7 signatures were similarly expressed in the iBlastoids (Extended Data Fig. 5b). Together, these results confirm the presence of EPI-, TE-, and PE-like cells in the iBlastoids. Finally, to facilitate the exploration of the iBlastoids and blastocyst datasets[12,13], we developed an interactive online tool (http://blastoid.ddnetbio.com/).

Transcriptomic similarity to blastocysts

Unsupervised clustering analysis identified seven cell clusters, which were assigned as EPI, TE, or PE clusters. Three remaining clusters did not have a clear identity a priori and seemed to have an intermediate signature (clusters IM1–3) (Fig. 2c, Extended Data Fig. 6a, b, Supplementary Table 3). Correlation analysis with day 21 reprogrammed cells showed that only IM1 correlated with one cluster at day 21, suggesting that these cells might already be present at that point (Extended Data Fig. 6c, d). Similarly, mouse blastoids have been reported to contain cell clusters with an intermediate or undefined signature[9,10,24]. We also observed a small cluster of cells representing non-reprogrammed refractory cells (Extended Data Fig. 5c, d) which did not contribute to the iBlastoids and were therefore excluded from downstream analysis.

To determine the transcriptional similarity between iBlastoids and blastocysts, we integrated the iBlastoid scRNA-seq data with two additional scRNA-seq datasets obtained from human blastocysts that were reported previously by the groups of Lanner[13] and Niakan[12]. UMAP analyses revealed a high concordance between cells of the blastocysts and iBlastoids (Fig. 2d, Extended Data Fig. 6e–g, Supplementary Table 4). Notably, cells from the EPI-, TE-, and PE-like clusters in the iBlastoids overlapped with their EPI, TE, and PE counterparts from blastocysts (Fig. 2d). Moreover, principal component analysis (PCA) indicated that EPI-, TE-, and PE-like cells of the iBlastoids showed a closer transcriptional resemblance to the EPI, TE, and PE cells of blastocysts than to pluripotent stem (PS), trophoblast stem (TS) and naive extraembryonic endoderm (nEnd) cell lines (Extended Data Fig. 6h). Unsupervised clustering confirmed that PE-, EPI-, and TE-like cells from iBlastoids were clustered together with cells of the PE, EPI, and TE, respectively, from blastocysts (Extended Data Fig. 7a, b). On this note, some blastocyst TE cells[13] clustered together with cells of the IM1 iBlastoid cluster. Furthermore, correlation analysis with the annotated EPI, TE, and PE cells from E5–7 blastocysts[12,13] revealed a high correlation (about 0.9) with their iBlastoid cluster counterparts (Fig. 2e, Extended Data Fig. 7c).

TE cells in human blastocysts specify into the polar and mural TE during embryogenesis[13]. Examination of the polar and mural TE signatures[13] (Extended Data Fig. 7d–f, Supplementary Table 5) in the iBlastoids showed that iBlastoids have two distinct populations of TE-like cells, one expressing a higher polar TE-related signature and the other expressing a mural TE signature (Fig. 2f, g, Extended Data Fig. 7g). Immunostaining analysis of the polar marker CCR7[13] also suggested higher expression at the polar side of the iBlastoids (Extended Data Fig. 7h, i).

Derivation of stem cells from iBlastoids

PS and TS cells are derived in vitro from blastocysts[2,25]. Similarly, we derived naive PS cells, primed PS cells and TS cells from iBlastoids (referred to as naive bPSCs, primed bPSCs and bTSCs, respectively; Fig. 3a, Extended Data Fig. 8a) and maintained them for more than 25 passages. Naive bPSCs showed a typical dome-shaped morphology and positive staining for KLF17 and NANOG, and expressed naive pluripotency-associated markers (*DPPA3*, *KLF17*, *DNMT3L*, *DPPA5*, *SUSD2*, and *KLF5*). Primed bPSCs exhibited flattened epithelial morphology and positive staining for TRA-160 and NANOG, and expressed primed pluripotency-associated markers (*ZIC2*, *ZIC3*, *NLGN4X*, *OTX2*, *CD24*, and *SALL2*). bTSCs showed the expected cobblestone-like morphology and were positive for KRT7 (a pan-trophoblast marker) and GATA2, as well as expressing trophoblast-associated genes such as *GATA2*, *GATA3*, *TFAP2C*, *KRT7*, *TP63*, and *TEAD4* (Fig. 3b–g, Extended Data Fig. 8b, c). Notably, we demonstrated that naive and primed bPSCs have trilineage differentiation potential (Extended Data Fig. 8d–h). Moreover, we confirmed that bTSCs can give rise to extravillous cytotrophoblasts (EVTs) and syncytiotrophoblasts (STs) (Extended Data Fig. 8i–n). These results suggest that iBlastoids have the capability to give rise to the major stem cell types that can be derived from blastocysts.

iBlastoids model implantation in vitro

To further define the identity of the structures that we obtained and evaluate whether they could recapitulate in vitro some aspects of early development, we adapted a published in vitro attachment assay previously used to model the peri- and early post-implantation processes in human blastocysts[14,21] (Extended Data Fig. 9a). The International Consensus and National law for culturing human embryos establishes that embryos obtained by IVF can be cultured up to 14 days post-fertilization and/or the formation of a primitive streak, whichever is first (see Methods). The applicability of the '14-day rule' to in vitro models of early development that are not derived by fertilization is not clear. Given the unclear nature of the structures we were analysing and to be cautious, we monitored our cultures meticulously and performed quantitative PCR with reverse transcription (qRT–PCR) for several key primitive streak markers[15,26] every 24 h and we only cultured the iBlastoids for up to 5 days. We did not observe upregulation of *TBXT*, *EOMES*, or *MIXL1*, or any morphological changes indicative of gastrulation[26] (Extended Data Figure 9b). Furthermore, to stringently adhere to the above parameters, we performed all subsequent attachment experiments for a total of 4.5 days.

Most (more than 90%) of the iBlastoids attached within 24 h, increased in size, flattened, and progressed to form an outgrowth,

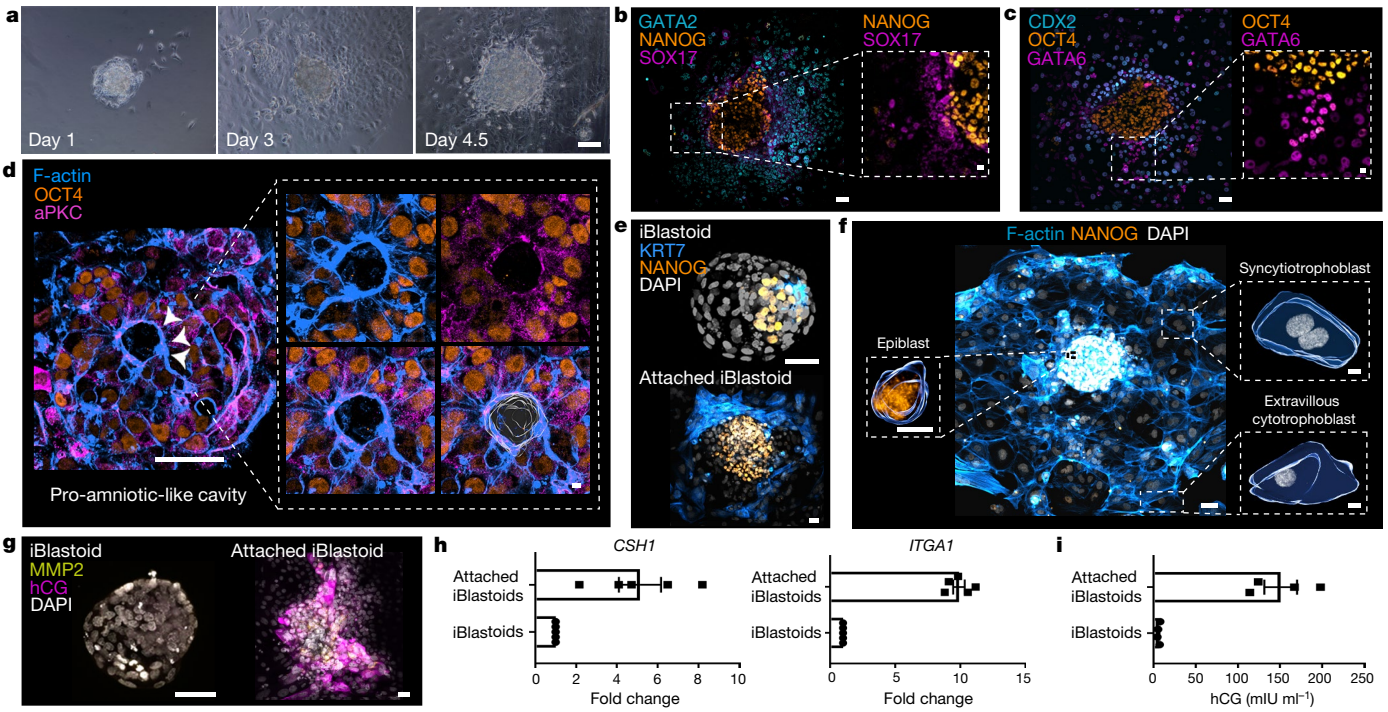


Fig. 4 | Modelling human in vitro implantation using iBlastoids. **a**, Representative phase-contrast images of iBlastoids at days 1, 3 and 4.5 of attachment; $n = 5$. **b**–**d**, Co-immunostaining of iBlastoids for GATA2, NANOG and SOX17 (**b**), CDX2, OCT4 and GATA6 (**c**) and F-actin, OCT4 and aPKC (**d**; arrowheads, pro-amniotic-like cavity); $n = 2$. Dashed areas expanded to right. **e**, Co-immunostaining of attached and unattached iBlastoids for KRT7 and NANOG; $n = 4$. **f**, Co-immunostaining of iBlastoids for F-actin and NANOG with expanded views of epiblast-like cell and putative ST and EVT; $n = 2$. **g**, Co-immunostaining of attached and unattached iBlastoids for MMP2 and hCG; $n = 2$. **h**, qRT–PCR analysis of ST marker CSH1 (left) and EVT marker ITGA1 (right) in attached and unattached iBlastoids, expressed as fold change over iBlastoids; mean ± s.e.m., $n = 5$. **i**, hCG protein level detected by enzyme-linked immunosorbent assay (ELISA) using conditioned medium collected from attached and unattached iBlastoids; mean ± s.e.m., $n = 4$. Scale bars, main images, 50 µm; expanded views, 10 µm.

similar to what has been reported in human blastocysts[14,21] (Fig. 4a). After attachment, the number of cells expressing NANOG or OCT4 and SOX2 increased, indicating expansion of the iBlastoid EPI-like lineage (Extended Data Fig. 9c, d), as observed in human blastocysts[14,21]. CDX2- and GATA2-positive cells also spread upon attachment (Extended Data Fig. 9c, d), which indicated outgrowth of the TE-like lineage in the attached iBlastoids. Although many PE-like cells still co-stained with TE markers (GATA2 or CDX2), we noticed SOX17- and GATA6-positive cells localizing to the perimeter of the NANOG- or OCT4-positive EPI-like compartments[21,27] (Fig. 4b, c, Extended Data Fig. 9e, f).

It has previously been reported that, upon in vitro attachment, EPI cells of human blastocysts polarized and formed a lumen known as the pro-amniotic cavity[14]. Immunostaining for F-actin, OCT4, and aPKC indicated a similar central lumen (marked by F-actin and aPKC) within the EPI-like compartment of about 20–30% of attached iBlastoids (Supplementary Table 6), surrounded by radially organized OCT4-positive cells (Fig. 4d, Extended Data Fig. 9g, Supplementary Video 3). We observed the polarization of EPI-like cells and the emergence of a pro-amniotic-like cavity on day 3 of attachment (Extended Data Fig. 10a).

We found high-intensity, filamentous KRT7 staining of the TE-like cells in attached iBlastoids, in contrast to the dull and limited KRT7 staining in iBlastoids before attachment culture (Fig. 4e, Extended Data Fig. 10b). This suggests a TE-like cellular state transition to a trophoblast-like lineage, as has been reported in blastocysts[15,21,28]. Notably, the nuclear volume of cells surrounding the EPI-like compartment was higher than that of cells in the EPI-like compartment, which is indicative of trophoblast-like cells[14] (Fig. 4f, Extended Data Fig. 10c, Supplementary Video 4). We also found cells that morphologically resembled STs and EVTs, as demonstrated by the multi-nucleated phenotype and hCG staining (indicating ST-like cells) and spindle-like morphology (indicating EVT-like cells)[14,28], in the periphery of the attached iBlastoids (Fig. 4f, g, Extended Data Fig. 10d). Although we found no expression of HLA-G in the EVT-like cells, we detected the EVT marker MMP2 in a smaller number of cells, which is in agreement with the development of EVTs later during the implantation stage[14,28] (Fig. 4g, Extended Data Fig. 10d, e). In addition, qRT–PCR analysis showed that the ST markers *CSH1*, *SDC1*, and *HSD3B1* and the EVT markers *ITGA1*, *ITGA5*, and *FN1*[25] were upregulated after attachment, indicating that the attached iBlastoid TE-like cells could differentiate into ST- and EVT-like states (Fig. 4h, Extended Data Fig. 10f). On this note, we detected a tenfold increase in the amount of hCG in the medium 4.5 days after attachment (Fig. 4i). Finally, we also characterized the iBlastoids using the in vitro fertilization blastocyst quality criteria score[29] (good, 1; fair, 2; poor, 3), according to which iBlastoids were graded as good or fair with an average score of 1.75 for ICM and 1.67 for TE (Extended Data Fig. 10g–i, Supplementary Table 7).

Discussion

We have presented an unanticipated approach for generating a 3D in vitro model system of the human blastocyst; unlike the approaches used to generate mouse blastoids[5–10], our approach was uncovered by studying the behaviour of reprogramming intermediates rather than the assembly of previously obtained stem cell lines. As iBlastoids are generated from somatic cells, without destruction of a human embryo, we anticipate that this will facilitate the wider acceptance of iBlastoids as a model, as has occurred for iPS cells in the past[3]. We show that iBlastoids can recapitulate several aspects of the human blastocyst: (1) iBlastoids have TE- and EPI-like layers, with cavitation, and model the spatial

positioning of cell types within a human blastocyst (Fig. 1); (2) iBlastoid cells are transcriptionally similar to cells of human blastocysts (Fig. 2); and (3) they can also be used to model in vitro some of the key functional characteristics of human blastocysts (Figs. 3, 4). By identifying these three main aspects of blastocyst biology in iBlastoids, we conclude that iBlastoids can be considered as an in vitro model of the blastocyst; however, they should not be considered as an equivalent to human blastocysts. For instance, they do not have a zona pellucida, and although we found PE-like cells within the ICM of iBlastoids, the structures seem to lack a defined PE cell layer. Moreover, the developmental potential of iBlastoids as a model for primitive streak formation and gastrulation remains to be determined, and will require an international conversation on the applicability of the 14-day rule to iBlastoids. Nevertheless, our data show that iBlastoids represent an accessible, scalable, and tractable model system that will be valuable for many applications in basic research and translational approaches. For example, iBlastoids provide an in vitro platform for modelling early cell-fate transitions during embryogenesis. Furthermore, we expect that iBlastoids could challenge the field to refine our existing understanding of human pluripotency by allowing us to re-establish the coordinated interactions of the TE and the PE with the EPI, which have remained inaccessible until now. Given that iBlastoids with specific genetic loads can be generated, this will allow studies of early developmental diseases and screening for treatments, and as such has tremendous potential for understanding infertility and early pregnancy loss[4]. iBlastoids could also serve as a platform for toxicity and viral susceptibility screens, as well as enabling gene therapy techniques. In summary, iBlastoids represent an opportunity to model, in vitro, the pre-implantation blastocyst and peri-implantation stages of human embryogenesis.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-021-03372-y.

1. Rossant, J. & Tam, P. P. L. New insights into early human development: lessons for stem cell derivation and differentiation. *Cell Stem Cell* **20**, 18–28 (2017).
2. Thomson, J. A. et al. Embryonic stem cell lines derived from human blastocysts. *Science* **282**, 1145–1147 (1998).
3. Takahashi, K. et al. Induction of pluripotent stem cells from adult human fibroblasts by defined factors. *Cell* **131**, 861–872 (2007).
4. Fu, J., Warmflash, A. & Lutolf, M. P. Stem-cell-based embryo models for fundamental research and translation. *Nat. Mater*. **20**, 132–144 (2021).
5. Harrison, S. E., Sozen, B., Christodoulou, N., Kyprianou, C. & Zernicka-Goetz, M. Assembly of embryonic and extraembryonic stem cells to mimic embryogenesis in vitro. *Science* **356**, eaal1810 (2017).
6. Sozen, B. et al. Self-assembly of embryonic and two extra-embryonic stem cell types into gastrulating embryo-like structures. *Nat. Cell Biol*. **20**, 979–989 (2018).
7. Rivron, N. C. et al. Blastocyst-like structures generated solely from stem cells. *Nature* **557**, 106–111 (2018).
8. Zhang, S. et al. Implantation initiation of self-assembled embryo-like structures generated using three types of mouse blastocyst-derived stem cells. *Nat. Commun*. **10**, 496 (2019).
9. Sozen, B. et al. Self-organization of mouse stem cells into an extended potential blastoid. *Dev. Cell* **51**, 698–712 (2019).
10. Li, R. et al. Generation of blastocyst-like structures from mouse embryonic and adult cell cultures. *Cell* **179**, 687–702 (2019).
11. Liu, X. et al. Reprogramming roadmap reveals route to human induced trophoblast stem cells. *Nature* **586**, 101–107 (2020).
12. Blakeley, P. et al. Defining the three cell lineages of the human blastocyst by single-cell RNA-seq. *Development* **142**, 3613 (2015).
13. Petropoulos, S. et al. Single-cell RNA-seq reveals lineage and X chromosome dynamics in human preimplantation embryos. *Cell* **165**, 1012–1026 (2016).
14. Shahbazi, M. N. et al. Self-organization of the human embryo in the absence of maternal tissues. *Nat. Cell Biol*. **18**, 700–708 (2016).
15. Xiang, L. et al. A developmental landscape of 3D-cultured human pre-gastrulation embryos. *Nature* **577**, 537–542 (2020).
16. Qin, H. et al. YAP induces human naive pluripotency. *Cell Rep*. **14**, 2301–2312 (2016).
17. Liu, L. et al. An integrated chromatin accessibility and transcriptome landscape of human pre-implantation embryos. *Nat. Commun*. **10**, 364 (2019).
18. Durruthy-Durruthy, J. et al. Spatiotemporal reconstruction of the human blastocyst by single-cell gene-expression analysis informs induction of naive pluripotency. *Dev. Cell* **38**, 100–115 (2016).
19. Fogarty, N. M. E. et al. Genome editing reveals a role for OCT4 in human embryogenesis. *Nature* **550**, 67–73 (2017).
20. Niakan, K. K. & Eggan, K. Analysis of human embryos from zygote to blastocyst reveals distinct gene expression patterns relative to the mouse. *Dev. Biol*. **375**, 54–64 (2013).
21. Deglincerti, A. et al. Self-organization of the in vitro attached human embryo. *Nature* **533**, 251–254 (2016).
22. Roode, M. et al. Human hypoblast formation is not dependent on FGF signalling. *Dev. Biol*. **361**, 358–363 (2012).
23. Kovacic, B., Vlaisavljevic, V., Reljic, M. & Cizek-Sajko, M. Developmental capacity of different morphological types of day 5 human morulae and blastocysts. *Reprod. Biomed. Online* **8**, 687–694 (2004).
24. Posfai, E. et al. Evaluating totipotency using criteria of increasing stringency. *Nat. Cell Biol*. **23**, 49–60 (2021).
25. Okae, H. et al. Derivation of human trophoblast stem cells. *Cell Stem Cell* **22**, 50–63 (2018).
26. Tyser, R. C. V., Mahammadov, E., Nakanoh, S. & Vallier, L. A spatially resolved single cell atlas of human gastrulation. Preprint at https://doi.org/10.1101/2020.07.21.213512 (2020).
27. Chen, D. et al. Human primordial germ cells are specified from lineage-primed progenitors. *Cell Rep*. **29**, 4568–4582 (2019).
28. West, R. C. et al. Dynamics of trophoblast differentiation in peri-implantation-stage human embryos. *Proc. Natl Acad. Sci. USA* **116**, 22635–22644 (2019).
29. Alpha Scientists in Reproductive Medicine and ESHRE Special Interest Group of Embryology. The Istanbul consensus workshop on embryo assessment: proceedings of an expert meeting. *Hum. Reprod*. **26**, 1270–1283 (2011).

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Methods

Ethics statement

This research was performed first under the oversight and approval of the Monash University Human Research Ethics Committee (MUHREC 2020-22909-39935) and later under MUHREC 2020-27147-51995, which served as the specialized human embryo research oversight (EMRO). In the first human research ethics committee review (MUHREC 2020-22909-39935), the MUHREC deemed that EMRO was not required given that 1) the original research goal was the functional and molecular characterization of human fibroblasts undergoing reprogramming using different culture conditions (there was no intention to make blastocyst-like structures from reprogramming intermediates, neither was it foreseeable); and 2) the research was based on the reprogramming of fibroblasts and did not involve the use of embryonic stem cells, embryos, embryo materials or gametes, nor performed in vitro fertilization. Once the molecular and functional data demonstrated that blastocyst-like structures were generated, the MUHREC and the Australian National Health and Medical Research Council (NHMRC) Embryo Research Licensing Committee (ERLC) were contacted. The NHMRC-ERLC advised that processing and analysing the inanimate material and data collected to that point did not require an embryo research licence, and so could continue. However, the Committee advised to stop the production of iBlastoids, while the Committee considered the regulatory path forward for future iBlastoid work in Australia. A new ethics application process was also initiated to specifically cover the generation, molecular and functional characterization of iBlastoids and this was approved (MUHREC 2020-27147-51995) by an EMRO process at Monash University bolstered by the expertise of embryo and ethics specialists in accordance with the ISSCR 2016 guidelines for research involving the experimental generation of embryo-like structures (2.1.3.2 Category 2 (g)).

The national and international consensus and the 2016 ISSCR Guidelines do not permit research that cultures human embryos or embryo-like structures beyond 14 days post-fertilization and/or formation of the primitive streak, whichever occurs first[30,31]. Given that the applicability of the '14-day rule' to human iBlastoids is unclear, as the iBlastoids are derived from adult donor fibroblasts and not via fertilization, making the time frame of 14 days ambiguous, we cultured the iBlastoids for the minimal time possible and stopped all experiments before evidence of a primitive streak. We performed the attachment assay for only a maximum of five additional days, and, to rule out morphological and molecular evidence of primitive streak formation, we monitored our cultures meticulously and performed qRT–PCR for several key primitive streak markers[15,26] every 24 h during the 5-day iBlastoid attachment culture. We never detected morphological or molecular evidence of the appearance of primitive streak (Extended Data Fig. 9b). In order to stringently adhere to the above parameters, we terminated the experiments 4.5 days after iBlastoid attachment. All experiments were conducted at Monash University and non-Monash investigators were not involved in any work that required institutional ethics review.

Cell culture media

Fibroblast medium. Fibroblast medium consisted of Dulbecco's modified Eagle's medium (DMEM; ThermoFisher), 10% fetal bovine serum (FBS; ThermoFisher), 1% nonessential amino acids (ThermoFisher), 1 mM GlutaMAX (ThermoFisher), 1% penicillin–streptomycin (ThermoFisher), 55 μM 2-mercaptoethanol (ThermoFisher) and 1 mM sodium pyruvate (ThermoFisher).

in vitro culture 1 (IVC1) medium. IVC1 medium[14] consisted of advanced DMEM/F-12 (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 2 mM L-glutamine (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), 20% FBS (ThermoFisher), 25 μM *N*-acetyl-L-cysteine (Sigma), 8 nM β-oestradiol (Sigma) and 200 ng/ml progesterone (Sigma).

IVC2 medium. IVC2 medium[14] consisted of advanced DMEM/F-12 (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 2 mM L-glutamine (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), 30% knockout serum replacement (KSR, ThermoFisher), 25 μM *N*-acetyl-L-cysteine (Sigma), 8 nM β-oestradiol (Sigma) and 200 ng/ml progesterone (Sigma).

Human iBlastoid medium. Human iBlastoid medium consisted of a 2:1:1 mixture of IVC1 medium, iBlastoid basal medium 1 (50:50 mixture of DMEM/F-12 (ThermoFisher) and neurobasal medium (ThermoFisher), supplemented with 2 mM L-glutamine (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% N2 supplement (ThermoFisher), 1% B27 supplement (ThermoFisher), and 1% penicillin–streptomycin (ThermoFisher)) and iBlastoid basal medium 2 (DMEM/F-12, GlutaMAX (ThermoFisher) supplemented with 0.3% bovine serum albumin (BSA; Sigma), 0.2% FBS (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), and 1.5 μg/ml L-ascorbic acid (Sigma)) supplemented with 2 μM CHIR99021 (Miltenyi Biotec), 0.5 μM A83-01 (Sigma), 1 μM SB431542 (Selleckchem), 0.8 mM valproic acid (VPA; Sigma), 50 ng/ml EGF (Peprotech) and 10 ng/ml BMP4 (Miltenyi Biotec).

Naive t2iLGoY medium. Naive t2iLGoY medium[32] consisted of a 50:50 mixture of DMEM/F-12 (ThermoFisher) and neurobasal medium (ThermoFisher), supplemented with 2 mM L-glutamine (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% N2 supplement (ThermoFisher), 1% B27 supplement (ThermoFisher), 1% penicillin–streptomycin (ThermoFisher), 10 ng/ml human leukaemia inhibitory factor (LIF, made in house), 250 μM L-ascorbic acid (Sigma), 10 μg/ml recombinant human insulin (Sigma), 1 μM PD0325901 (Miltenyi Biotec), 1 μM CHIR99021 (Miltenyi Biotec), 2.5 μM Gö6983 (Tocris), and 10 μM Y-27632 (ROCK inhibitor; Selleckchem).

Essential 8 medium. Essential 8 medium consisted of essential 8 basal medium (ThermoFisher) supplemented with 50× essential 8 supplement (ThermoFisher) and 1% penicillin–streptomycin (ThermoFisher).

TSC medium. TSC medium[25] consisted of DMEM/F-12, GlutaMAX (ThermoFisher) supplemented with 0.3% BSA (Sigma), 0.2% FBS (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), 1.5 μg/ml L-ascorbic acid (Sigma), 5 μM Y27632 (ROCK inhibitor; Selleckchem), 2 μM CHIR99021 (Miltenyi Biotec), 0.5 μM A83-01 (Sigma), 1 μM SB431542, 50 ng/ml EGF (Peprotech) and 0.8 mM VPA (Sigma).

iBlastoid generation

The experimental design, materials, and reagents are described in the Life Sciences Reporting Summary. All cell lines used in this study were authenticated and tested for mycoplasma as described in the Reporting Summary. A detailed step-by-step protocol for generation of iBlastoids is available on Protocol Exchange[33]. Primary human adult dermal fibroblasts (HDFa) from three female donors were obtained from ThermoFisher (catalogue number C-013-5C; lot 1029000 for 38F, lot 1528526 for 55F and lot 1569390 for 32F). Cells were recovered and plated in medium 106 (ThermoFisher) supplemented with low-serum growth supplement (LSGS) (ThermoFisher) for expansion. Human somatic cell reprogramming was performed as previously described[11,34] to obtain the day 21 reprogrammed cells. In brief, reprogramming of human fibroblasts was conducted using a CytoTune-iPS 2.0 Sendai reprogramming kit according to the manufacturer's instructions (ThermoFisher, lot 2170052). Primary HDFa were seeded at a density of about $5–10 \times 10^4$ cells per well on a 12-well

plate in fibroblast medium. As shown in Fig. 1a, cells were transduced with Sendai viruses in fibroblast medium at multiplicities of infection (MOI) as follows: *KOS* MOI = 5, *c-MYC* MOI = 5, *KLF4* MOI = 6. Medium was replaced every other day starting from day 1 following transduction and every day from day 8 onwards. On day 21 of reprogramming, cells were dissociated and seeded at a density of 1.2×10^5 cells per well onto a 24-well AggreWell plate (Stem Cell Technologies) in human iBlastoid medium supplemented with 10 μM Y-27632 (ROCK inhibitor, Selleckchem) according to the manufacturer's instructions. The cells were cultured in incubators at 37 °C, 5% O_2 and 5% CO_2. After 24 h, the cells were replenished with fresh human iBlastoid medium without ROCK inhibitors. On day 6 of iBlastoid formation in the AggreWell, the iBlastoids were collected and selected for subsequent analysis or in vitro attachment assay. To quantify the efficiency of iBlastoid formation, 628 randomly selected iBlastoids from three donors were counted and examined for the number of structures with a typical blastocyst-like morphology and a blastocoel cavity. Details of the culture media used in the generation of iBlastoids are summarized in the 'Cell culture media' section.

In vitro attachment assay

The in vitro attachment assay was performed by adapting a previously published protocol[14,21,28,35]. In brief, iBlastoids derived were collected and sedimented for 5–10 min to remove the majority of cell debris within the supernatant. The iBlastoids were then transferred onto optical-grade tissue culture plates (Eppendorf) and cultured in IVC1 medium at 37 °C, 20% O_2 and 5% CO_2 in incubators. On day 2 of the attachment assay, the culture medium was switched to IVC2 medium. iBlastoids were cultured up to day 4.5 in the attachment assay and collected for analysis. The experiments were performed by strictly following the guidelines and protocols described in the Ethics Statement. Details of the culture media used in the in vitro attachment assay are summarized in the 'Cell culture media' section.

Derivation of stem cells from iBlastoids

For the derivation of naive bPSCs, iBlastoids were collected, dissociated and re-seeded onto irradiated mouse embryonic fibroblast (iMEF) feeders[36] in human iBlastoid medium supplemented with 10 μM Y-27632 (ROCK inhibitor, Selleckchem). For the derivation of primed bPSCs, dissociated cells were seeded onto 5 μg/ml vitronectin (ThermoFisher)-coated plates in human iBlastoid medium supplemented with 10 μM Y-27632 (ROCK inhibitor, Selleckchem). For the derivation of bTSCs, dissociated cells were seeded onto 5 μg/ml collagen IV (Sigma)-coated plates in human iBlastoid medium supplemented with 10 μM Y-27632 (ROCK inhibitor, Selleckchem). After 24 h, the cell culture conditions were switched to naive t2iLGoY medium, essential 8 medium and TSC medium, respectively.

For the maintenance of naive bPSCs, the cells were passaged every 5–6 days at a 1:10–1:20 ratio using Accutase onto iMEF feeders with daily medium replacement. For the maintenance of primed bPSCs, the cells were passaged every 5–6 days at a 1:15–1:25 ratio using 0.5 mM EDTA (ThermoFisher) onto 5 μg/ml vitronectin (ThermoFisher)-coated plates with daily medium replacement. The medium was supplemented with 10 μM Y-27632 (ROCK inhibitor, Selleckchem) during the first day of primed bPSC passaging. For the maintenance of bTSCs, the cells were passaged every 4–6 days at a 1:4–1:10 ratio using TrypLE Express (ThermoFisher) onto 5 μg/ml collagen IV (Sigma-Aldrich)-coated plates with medium replacement every other day.

For the assessment of pluripotency in naive and primed bPSCs, a TaqMan hPSC Scorecard Assay (ThermoFisher) and trilineage differentiation assay were performed. Previous reports indicated that naive PSCs need to be transitioned into the primed state to gain trilineage differentiation potential[34,37], so the naive bPSCs were converted into primed bPSCs (marked with # in 'Statistics and reproducibility' and Extended Data Figures) for these downstream assays of pluripotency. To perform the TaqMan hPSC Scorecard Assay, primed bPSCs at 70% confluency were gently scraped off using a cell scraper and cultured in DMEM/F-12, GlutaMAX (ThermoFisher) supplemented with 20% KSR (ThermoFisher), 1% non-essential amino acids (ThermoFisher) and 55 μM 2-mercaptoethanol (ThermoFisher) for embryoid body (EB) formation on ultra-low-binding attachment plates (Corning), with medium replacement every other day. The EBs were harvested on day 7 of incubation and qRT–PCR using the TaqMan hPSC Scorecard Panel was performed according to the manufacturer's instructions. The trilineage differentiation assay of primed bPSCs was performed using a STEMdiff Trilineage Differentiation Kit (Stem Cell Technologies) according to the manufacturer's instructions.

Differentiation of bTSCs into STs and EVTs was performed and modified as previously described[25]. For the differentiation of bTSCs into STs, bTSCs were seeded at a density of 3.75×10^4 cells per well onto a 24-well plate pre-coated with 2.5 μg/ml collagen IV (Sigma) and cultured in 500 μl ST differentiation medium (DMEM/F-12, GlutaMAX (ThermoFisher) supplemented with 0.3% BSA (Sigma), 4% KSR (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), 2.5 μM Y27632 (Selleckchem) and 2 μM forskolin (Selleckchem)). Medium was replaced on day 3 of differentiation, and cells were analysed on day 6. The fusion index was calculated by using the number of nuclei counted in the syncytia minus the number of syncytia, then divided by the total number of nuclei counted. For the differentiation of bTSCs into EVTs, bTSCs were seeded at a density of 3.4×10^4 cells per well onto a 24-well plate pre-coated with 1 μg/ml collagen IV (Sigma) and cultured in 500 μl EVT differentiation medium (DMEM/F-12, GlutaMAX (ThermoFisher) supplemented with 0.3% BSA (Sigma), 4% KSR (ThermoFisher), 1% ITS-X supplement (ThermoFisher), 0.1 mM 2-mercaptoethanol (ThermoFisher), 0.5% penicillin–streptomycin (ThermoFisher), 2.5 μM Y27632 (Selleckchem), 100 ng/ml hNRG1 (Cell Signaling), 7.5 μM A83-01 (Sigma) and 2% Matrigel (Corning). On day 3 of differentiation, the medium was replaced with EVT differentiation medium without hNRG1, and Matrigel (Corning) was added to a final concentration of 0.5%. On day 6 of differentiation, EVT differentiation medium was replaced without hNRG1 (Cell Signaling) and KSR (ThermoFisher), and Matrigel (Corning) was added to 0.5% final concentration. The cells were cultured for an additional 2 days before analyses were performed.

Immunofluorescence staining

iBlastoids and cells were fixed in 4% paraformaldehyde (PFA, Sigma) for 30 min at room temperature, permeabilized with 0.5% Triton X-100 (Sigma) in DPBS (ThermoFisher) for 30 min at room temperature and blocked with blocking buffer (3% bovine serum albumin (BSA) (Sigma) + 0.1% Tween-20 (Sigma) in Dulbecco's phosphate-buffered saline (DPBS; ThermoFisher)) for 3–4 h at 4 °C. All antibodies used in this study are described in Supplementary Table 9 and the Reporting Summary linked to this paper. For example, primary antibodies used: rabbit anti-NANOG polyclonal (1:100, Abcam), mouse anti-CDX2 IgG1 (1:50, Abcam) prepared in a blocking buffer. Primary antibody incubation was conducted overnight at 4 °C on shakers followed by room temperature incubation with secondary antibodies (1:500 in blocking buffer) for 3 h. Secondary antibodies used in this study were goat anti-rabbit IgG AF555 (1:500, ThermoFisher) or goat anti-rabbit IgG AF647 (1:500, Invitrogen) for NANOG, and goat anti-mouse IgG AF488 (1:400, ThermoFisher) or goat anti-mouse IgG AF488 (1:500, ThermoFisher) for CDX2 (Supplementary Table 9). After labelling, iBlastoids and cells were stained with 4′,6-diamidino-2-phenylindole, dihydrochloride (DAPI, ThermoFisher) at a concentration of 5 μg/ml in a blocking buffer for 1 h. Images were taken using a DMi8 inverted microscope (Leica), an SP8 inverted confocal microscope (Leica) or an LSM780 multiphoton confocal inverted microscope (Zeiss). iBlastoid staining was compared with blastocyst staining as previously published[12–15,19–23,27,28,35,38–41].

Confocal imaging and image analysis

Immunostained iBlastoids were imaged using an SP8 inverted confocal microscope (Leica) or a laser scanning confocal microscope (LSM 780, Zeiss) with a water UV-VIS-IR Apochromat 40× 1.2 NA objective and highly sensitive avalanche photodiode light detectors of the Confocor 3 module (Zeiss). All images in this study were acquired using Motic Image Plus, Leica application suite X or ZEISS Efficient Navigation software. 3D visualizations of iBlastoids were performed using Imaris 9.5 software (Bitplane AG). The manual surface-rendering module was used for cell and iBlastoid segmentation. Final images were processed and assembled using Adobe Photoshop or ImageJ. Whole-well fluorescence images of day 21 reprogrammed cells were acquired using a DMi8 microscope (Leica) and quantified for different cell populations using QuPath[42].

Fluorescence-activated cell sorting (FACS)

iBlastoids were dissociated using TrypLE Express (ThermoFisher), and DPBS (ThermoFisher) supplemented with 2% FBS (ThermoFisher) and 10 µM Y-27632 (Selleckchem) was used for final resuspension of the samples. Dissociated cells were pelleted at 400*g* for 5 min and then resuspended in a final volume of 500 µl with propidium iodide (PI) (Sigma) added to a concentration of 2 µg/ml. Cell sorting was carried out with a 100-µm nozzle on an Influx instrument (BD Biosciences).

qRT–PCR

RNA was extracted from cells using RNeasy micro kit (Qiagen) or RNeasy mini kit (Qiagen) and QIAcube (Qiagen) according to the manufacturer's instructions. Reverse transcription was then performed using QuantiTect reverse transcription kit (Qiagen, catalogue no. 205311). Real-time PCR reactions were set up in duplicate using QuantiFast SYBR Green PCR Kit (Qiagen) and then carried out on the 7500 Real-Time PCR system (ThermoFisher) using LightCycler 480 software. Information regarding the qRT–PCR primers used in this study is available in Supplementary Table 8.

Mesoderm differentiation for positive control of primitive streak markers

A positive control for qRT–PCR of primitive streak markers was obtained by modifying a previously published mesoderm differentiation protocol[43]. In brief, human iPSCs were grown in E8 medium (ThermoFisher) and at 50% confluency the medium was replaced with a culture medium consisting of RPMI, GlutaMAX (ThermoFisher), 1% B27 supplement (ThermoFisher), 1% penicillin–streptomycin (ThermoFisher) and 5 µM CHIR99021 (Miltenyi Biotec). After 48 h, the differentiated cells were collected for qRT–PCR analysis, which showed high expression of the primitive streak markers *TBXT*, *EOMES* and *MIXL1* (Extended Data Fig. 9b).

hCG ELISA

Generation of iBlastoids and the in vitro attachment assay were performed as described in 'In vitro attachment assay'. The medium was collected for both iBlastoids (day 6) and attached iBlastoids (day 6 + 4.5) and stored at −80 °C. The hCG level in the medium was measured using an hCG ELISA kit (Abnova, ABNOKA4005) according to the manufacturer's instructions.

scRNA-seq of iBlastoids

For scRNA-seq experiments, we selected approximately 200 iBlastoids from two donors on the basis of morphological assessment. The selection criteria were: (1) presence of lumen; (2) presence of ICM-like aggregates within the lumen; and (3) similarity in size to blastocysts (Extended Data Fig. 4a). The selected iBlastoids were dissociated to obtain single-cell suspensions for FACS as described above. Cells subjected to FACS were sorted for PI-negative, non-debris live single cells for scRNA-seq (Supplementary Fig. 1). The collected cells were isolated, encapsulated and constructed using Chromium controller (10x Genomics) according to the manufacturer's instructions (Chromium Next GEM Single Cell 3′ Reagent Kit V3.3 User Guide). Sequencing was done on an Illumina NovaSeq 6000 using a paired-end (R1 28 bp and R2 87 bp) sequencing strategy and aiming for 20,000 read-pairs per cell. Chromium barcodes were used for demultiplexing and FASTQ files were generated from the mkfastq pipeline using the Cellranger program (v3.1.0, http://software.10xgenomics.com/single-cell/overview/welcome). Alignment and UMI counting were performed using Cellranger, which uses the STAR aligner[44] (v2.5.1b) to map 194,957,954 sequencing reads to a custom version of the Ensembl GRCh37.87 reference genome, which we expanded by sequences of the custom SENDAI *KLF4*, *MYC*, and SEV (*KOS*) vectors. Alignment could successfully map 93.4% of reads to the genome. This step resulted in 9,060 unique cell barcodes with an average of 21,519 reads per cell and a median of 2,906 genes.

iBlastoid scRNA-seq cell donor identification

To determine the identity of the individual donor of every cell, we used the Bayesian demultiplexing tool Vireo (v 0.3.2)[45]. In brief, we first piled up expressed alleles in the single-cell data using cellSNP (v 0.3.0) to generate a list of single nucleotide polymorphisms (SNPs) with a minimum allele frequency (using the argument minMAF) of 0.1 and minimum unique molecular identifiers (using the argument minCOUNT) of 20. Furthermore, cellSNP requires a reference list of human variants to call the SNPs and we used a pre-compiled list of SNPs from the 1000 Genome Project provided by the authors of Vireo (downloaded from https://sourceforge.net/projects/cellsnp/files/SNPlist/). In particular, the list of variants based on the hg19 genome with minor allele frequency >0.05, containing 7.4 million SNPs was used. Subsequently, Vireo was performed to demultiplex the single-cell library by separating the cells into two donor populations. Note that Vireo is only able to distinguish the cells with respect to the two donors, but is unable to assign the exact donor identity (32F or 38F) to each cell.

iBlastoid scRNA-seq cell calling and quality control

Quality control was first performed at the cell level. Cells with (1) evidence for both or neither cell line donor, (2) a low number of expressed genes (nGene), or (3) a high percentage of mitochondrial genes (pctMT) were discarded. Cutoffs of nGene < 1,300 and pctMT > 15 were applied to discard cells. Next, quality control was performed at the gene level. Genes were filtered if not present in at least 50 cells with at least 1 read each. All cutoffs were determined after investigating the distributions of each variable. After quality control, 6,858 cells and 14,224 genes remained for scRNA-seq.

iBlastoid scRNA-seq analysis

Analyses in the remainder of the section were conducted using R (v3.6)[46] with Seurat (v3.1.5)[47,48]. Bioinformatics plots were generated using ggplot2 (v3.3.1)[49], and heat maps with pheatmap (v1.0.12)[50]. The SCTransform function[51] in Seurat was used to scale and normalize the data. Following PCA, a UMAP was generated using 20 dimensions. Unsupervised clustering was performed using the FindClusters function with a resolution of 0.2, resulting in seven individual clusters. Genes that were differentially expressed between clusters (cluster markers) were identified with the FindAllMarkers function, using a Wilcoxon rank sum test and a minimum upregulation of 0.25 log-fold. Average overexpression of cell type signatures (EPI, TE, PE and refractory cells) was calculated with the AddModuleScore function and gene signatures published by Petropoulos et al.[13] and Liu et al.[11]. Cell types were assigned to clusters manually using canonical markers and the Petropoulos and Liu signatures as evidence. The remaining cell populations were labelled 'intermediate' (IM) and enumerated (1–3). Finally, cell cycle scores were calculated using the cyclone function[52] from the scran package, and cell phase assigned according to the highest probability.

Information regarding each single cell for the iBlastoid scRNA-seq dataset used in this study is available in Supplementary Table 1.

For the assessment of mural and polar TE signatures in the iBlastoid scRNA-seq dataset, the signatures were obtained from the Petropoulos dataset by selecting the top 100 differentially expressed polar and mural genes at E6 (sorted by absolute log fold-change) after filtering for $P < 0.01$. The subset of TE-like cells was then obtained from iBlastoids scRNA-seq for identification of variable genes and dimensionality reduction using SCTransform, RunPCA, and RunUMAP with ten dimensions. The mural and polar signatures were applied to the TE cell subset using AddModuleScore and the results were visualized as a UMAP plot and a histogram of binned signature scores along UMAP1. Lists of the mural and polar gene signatures are provided in Supplementary Table 5.

Integrated scRNA-seq analysis

Previously published single-cell datasets from Petropoulos et al.[13] and Blakeley et al.[12] were integrated with the iBlastoids datasets. First, cells from the NR cluster were removed as irrelevant to this integration from the iBlastoid data. Petropoulos's 1,529 cells were filtered for blastocyst cells, removing the pre-blastocyst stages to leave 1,096 E5–E7 EPI, TE, and PE cells. Petropoulos's data and Blakeley's 30 cells were processed using SCTransform. Both datasets were integrated into the iBlastoids data after identifying integration genes (using FindIntegrationAnchors and IntegrateData) using 4,000 anchor genes derived from the SCT assays. The integrated dataset has 7,861 cells, 23,308 genes, and 4,000 integrated genes. PCA and UMAP (20 dimensions) were used for dimensionality reduction and FindClusters to identify clusters (resolution 0.2). Cluster identities for five individual clusters were assigned manually using co-localization of cell identities from all three datasets as evidence, as well as marker and signature expression. Transcriptome correlation was calculated using the integrated gene expression values and Pearson correlation on mean values of gene expression across all cells of (1) the same original cell identity (cluster ids for iBlastoids; EPI, TE and PE for Blakeley; E5–7 EPI, TE and PE for Petropoulos), or (2) the same cell type (aggregates of all EPI, TE or PE cells and, additionally, IM cells for iBlastoids). Information regarding each single cell for the Petropoulos scRNA-seq dataset and the integrated iBlastoid, Petropoulos and Blakeley scRNA-seq dataset are available in Supplementary Tables 2, 4.

Bulk RNA-seq integration analysis

For Extended Data Fig. 6h, to integrate single-cell datasets with previously published bulk data including Blakeley et al.[12], Petropoulos et al.[13], Okae et al.[25] and Linneberg-Agerholm et al.[53], we proceeded as follows. Raw counts for cells of the same type (EPI, PE, TE; for Petropoulos's data, E5.EPI, E5.PE, E5.TE, and so on) were aggregated as a pseudo-bulk sample for each cell type. The samples for iBlastoids were further split into two replicates (for the two cell donors). $\log_2$[counts per million (CPM)] values were calculated for pseudo-bulk and bulk RNA-seq samples by adding pseudo-counts of 0.25 and normalization of library size, followed by $\log_2$ transformation. The full transcript length protocols (such as Blakeley et al.[12] and Petropoulos et al.[13]) were further normalized by gene length. We removed the batch effect between libraries using edgeR's removeBatchEffect function. The resulting normalized gene expression values of genes present in all datasets were reduced to principal components using the prcomp function. We visualized the reduction using ggplot2.

Statistics and reproducibility

The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

For the scRNA-seq of day 21 reprogrammed cells, the total number of cells used in the final analysis was 6,816 cells (Extended Data Figs. 1a, i, 6c, d). For Extended Data Fig. 6c, 6,816 cells across the cell clusters were: 1,624 cells for cluster 0, 1,013 cells for cluster 1, 753 cells for cluster 2, 719 cells for cluster 3, 634 cells for cluster 4, 620 cells for cluster 5, 532 cells for cluster 6, 501 cells for cluster 7, 329 for cluster 8, 91 cells for cluster 9. The scRNA-seq library of iBlastoids was obtained from $n = 2$ biological replicates (iBlastoids from two donors) pooled into the same library, yielding a total of 6,858 cells for the final analysis used in this study (Fig. 2a–c, Extended Data Figs. 4c–f, 5b–d, 6a, b). Specific information for the iBlastoid scRNA-seq library can be found in Supplementary Table 1. For Fig. 2c, 6,858 cells across the cell clusters were: 1,160 cells for EPI_iBlastoids, 1,416 cells for TE_iBlastoids, 1,659 cells for PE_iBlastoids, 1,800 cells for IM1_iBlastoids, 621 cells for IM2_iBlastoids, 79 cells for IM3_iBlastoids and 123 cells for NR_iBlastoids. For the scRNA-seq dataset of human blastocysts used in this study, 1,096 cells were adapted from the Petropoulos[13] dataset (Extended Data Figs. 4g, 5a) and 30 cells were adapted from the Blakeley[12] dataset for inclusion in final analysis. For the iBlastoids and human blastocyst integrated dataset, a total of 7,861 cells were used for final analysis (6,735 cells from the iBlastoids scRNA-seq dataset, 1,096 cells from the Petropoulos[13] dataset and 30 cells from the Blakeley[12] dataset) (Fig. 2d, e, Extended Data Figs. 6e–g, 7a–c). For Extended Data Fig. 7a, 7,861 cells across the cell clusters were: 1,715 cells for Int_EPI, 1,865 cells for Int_TE, 1,849 cells for Int_PE, 1,702 cells for Int_IM1, 730 cells for Int_IM2. For Fig. 2f, g and Extended Data Fig. 7g, a total of 1,416 cells from the iBlastoids_TE clusters were used for analysis. For Extended Data Fig. 7d–f, a total of 861 cells from the Petropoulos TE population were used for analysis. For Supplementary Table 3, the top marker genes among each cluster in the iBlastoid scRNA-seq dataset were obtained using Seurat's FindAllMarkers function with adjusted P values below 0.05. For Extended Data Fig. 6h, the samples incorporated in the bulk RNA-seq integration analysis were: 3 samples from the Blakeley[12] dataset (EPI, TE and PE); 9 samples from the Petropoulos[13] dataset (E6–E7 EPI, E6–E7 TE and E6–E7 PE); 6 samples from the iBlastoids dataset (EPI, TE and PE from 2 donors); 9 samples from the Linneberg-Agerholm et al.[53] dataset (triplicates of nES, triplicates of nEnd and triplicates of PrE); and 4 samples from the Okae et al.[25] dataset (two replicates of TS cells). Data presented in Figs. 1e, f, 4h, i and Extended Data Figs. 1f, n, 2c, 3c, 7i, 8c, k–n, 9b, 10f were plotted using GraphPad Prism. For Fig. 1b, representative phase contrast images of $n = 5$ biological replicates (5 independent experiments from 3 donors) are shown for iBlastoids formed in AggreWells. For Fig. 1c, representative phase contrast and immunostaining images of $n = 5$ biological replicates (5 independent experiments from 3 donors) are shown. For Fig. 1e, f, the x/y aspect ratio and projection area of iBlastoids were measured from $n = 18$ biological replicates (18 iBlastoids from 3 donors) and blastocysts were measured from $n = 8$ published references. Data are represented as mean ± s.e.m. For Fig. 1g, total cell number estimation was measured from $n = 14$ biological replicates (14 iBlastoids from 3 donors). Data are represented as a density distribution and a dashed line indicating mean cell number. For Fig. 1h, i, representative images are shown from immunostaining of $n = 5$ biological replicates (5 independent experiments from 3 donors). For Fig. 1j, representative images are shown from immunostaining of $n = 3$ biological replicates (3 independent experiments from 3 donors). For Fig. 1k–n, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Fig. 3b–g, representative phase contrast and immunostaining images of $n = 2$ biological replicates (2 cell lines from 2 donors) are shown. For Fig. 4a, representative phase contrast images of $n = 5$ biological replicates (5 independent experiments from 3 donors) are shown for the iBlastoids from day 1 to day 4.5 of the in vitro attachment assay. For Fig. 4b–d, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Fig. 4e, representative images are shown from immunostaining of $n = 4$ biological replicates (4 independent experiments from 2 donors). For Fig. 4f, g, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Fig. 4h, the relative expression of *CSH1* and *ITGA1* was measured in $n = 5$ independent experiments (from

2 donors) with technical replicates. Representative results are shown, data are presented as mean ± s.e.m. For Fig. 4i, the conditioned media of 4 independent experiments (from 2 donors) were measured in 2 technical replicates ($n = 4$ biological replicates × 2 technical replicates). Representative results are shown, data are presented as mean ± s.e.m.

For Extended Data Fig. 1b–d, representative images are shown from immunostaining of $n = 3$ biological replicates (from 3 donors). For Extended Data Fig. 1e, different existing cell populations in day 21 reprogrammed cells were assessed from $n = 3$ biological replicates (average proportion of 5 random regions each from 3 donors). For Extended Data Fig. 1f, the *x*-and *y*-axis diameters of iBlastoids were measured from $n = 18$ biological replicates (18 iBlastoids from 3 donors) and blastocysts were measured from $n = 8$ published references. Data are represented as mean ± s.e.m. For Extended Data Fig. 1g, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 1h, $n = 628$ biological replicates were assessed to quantify the proportion of iBlastoids with cavitation (628 iBlastoids from 3 donors, distributed into 15 independent rounds for quantification). For Extended Data Fig. 1j, representative phase contrast images of $n = 2$ biological replicates (2 independent experiments from 2 donors) were shown for refractory HDFa. For Extended Data Fig. 1k, the relative expression of *NANOG, SOX2, TWIST1* and *ZEB1* was measured in $n = 3$ independent experiments with technical replicates. Representative results are shown, data are represented as mean ± s.e.m. For Extended Data Fig. 2a, representative images are shown from immunostaining of $n = 5$ biological replicates (5 independent experiments from 3 donors). For Extended Data Fig. 2b, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 2c, the proportions of EPI-, PE- and TE-like cells out of total cells were assessed from $n = 5$ biological replicates (5 iBlastoids for EPI/PE and 5 iBlastoids for EPI/TE). Data are represented as mean ± s.e.m. For Extended Data Fig. 2d–f, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 3a, representative phase contrast images of $n = 5$ biological replicates (5 independent experiments from 3 donors) are shown for iBlastoid formation from day 1 to day 6. For Extended Data Fig. 3b, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 3c, the dmax/dmin ratio was measured from $n = 20$ biological replicates each for TE-like and EPI-like cells (20 cells each for TE-like and EPI-like cells of iBlastoids from 2 donors). For Extended Data Fig. 4a, representative phase contrast images of $n = 200$ biological replicates (200 iBlastoids from 2 donors) are shown for iBlastoids selected for scRNA-seq. For Extended Data Fig. 7h, representative images are shown from immunostaining of $n = 4$ biological replicates (4 iBlastoids). For Extended Data Fig. 7i, CCR7 fluorescence intensity was quantified on $n = 4$ biological replicates (4 iBlastoids) by measuring fluorescence intensity over five regions each from polar and mural TE of each iBlastoid. Data are represented as median with maxima and minima (whiskers). For Extended Data Fig. 8b, representative images are shown from immunostaining of $n = 2$ biological replicates (2 cell lines from 2 donors). For Extended Data Fig. 8c, the relative expression of each marker was measured in $n = 4$ independent experiments (from 2 donors) with technical replicates. Data are represented as mean ± s.e.m. For Extended Data Fig. 8d, e, representative phase contrast images of $n = 2$ biological replicates (2 cell lines from 2 donors) are shown for EBs from #primed bPSCs and primed bPSCs. For Extended Data Fig. 8f, the ScoreCard Assay analysis was performed on $n = 2$ biological replicates (from 2 donors) for EBs from #primed bPSCs and primed bPSCs. For Extended Data Fig. 8g–j, representative images are shown from immunostaining of $n = 2$ biological replicates (2 cell lines from 2 donors). For Extended Data Fig. 8k, l, the relative expression of each marker was measured in $n = 4$ independent experiments (from 2 donors) with technical replicates. Data are presented as mean ± s.e.m. For Extended Data Fig. 8m, quantification of fusion index was performed on $n = 12$ cell clusters counted randomly and independently across STs (from 2 donors) with similar results and representative results are shown. $P = 2.24 \times 10^{-22}$, data are represented as mean ± s.e.m., significance determined by two-tailed unpaired Student's *t*-test. For Extended Data Fig. 8n, the conditioned media of 2 independent experiments (from two donors) were measured in 2–3 technical replicates ($n = 2$ biological replicates × 2–3 technical replicates). Representative results are shown, data are presented as mean ± s.e.m. For Extended Data Fig. 9b, the relative expression of *TBXT*, *EOMES* and *MIXL1* was measured in $n = 6$ independent experiments (from 2 donors) with technical replicates. Data are presented as mean ± s.e.m. For Extended Data Fig. 9c, representative images are shown from immunostaining of $n = 5$ biological replicates (5 independent experiments from 3 donors). For Extended Data Fig. 9d–g, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 10a, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 10b, representative images are shown from immunostaining of $n = 4$ biological replicates (4 independent experiments from 2 donors). For Extended Data Fig. 10c–e, representative images are shown from immunostaining of $n = 2$ biological replicates (2 independent experiments from 2 donors). For Extended Data Fig. 10f, the relative expression of each marker was measured in $n = 5$ independent experiments (from 2 donors) with technical replicates. Representative results are shown, data are presented as mean ± s.e.m. For Extended Data Fig. 10h, phase-contrast images of iBlastoids included for scoring assessment are shown for $n = 24$ biological replicates (24 iBlastoids obtained from three donors). For Extended Data Fig. 10i, the ICM and TE grade were assessed from $n = 24$ biological replicates (24 iBlastoids from 3 donors) with the average grade presented.

Reporting summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

Data availability

All processed datasets are available at http://blastoid.ddnetbio.com/. Raw and processed next-generation sequencing datasets from iBlastoids were deposited at the Gene Expression Omnibus (GEO) repository under accession number GSE156596. Published human embryo datasets used in this study were obtained from Blakeley et al.[12] and Petropoulos et al.[13] under accession numbers GSE66507 and E-MTAB-3929 (ArrayExpress, https://www.ebi.ac.uk/arrayexpress/experiments/E-MTAB-3929/), respectively. For Extended Data Fig. 6h, datasets were obtained from Table S1 of Okae et al.[25] and Linneberg-Agerholm et al.[53] GSE138012 for the RNA-seq integration. There are no restrictions on data availability. Source data are provided with this paper.

Code availability

All data were analysed with standard programs and packages as detailed above. Scripts can be found at http://github.com/jibsch/iBlastoids_scripts.

30. Hyun, I., Munsie, M., Pera, M. F., Rivron, N. C. & Rossant, J. Toward guidelines for research on human embryo models formed from stem cells. *Stem Cell Rep.* **14**, 169–174 (2020).
31. Warnock—report of the committee of inquiry into human fertilisation and embryology. *Ir. Nurs. News* **5**, 7–8 (1985).
32. Guo, G. et al. Naive pluripotent stem cells derived directly from isolated cells of the human inner cell mass. *Stem Cell Rep.* **6**, 437–446 (2016).
33. Tan, J. P., Liu, X. & Polo, J. M. Generation of human blastocyst-like structures by somatic cell reprogramming. *Protoc. Exch*. https://doi.org/10.21203/rs.3.pex-1347/v1 (2021).

34. Liu, X. et al. Comprehensive characterization of distinct states of human naive pluripotency generated by reprogramming. *Nat. Methods* **14**, 1055–1062 (2017).
35. Zhou, F. et al. Reconstituting the transcriptome and DNA methylome landscapes of human implantation. *Nature* **572**, 660–664 (2019).
36. Paynter, J. M., Chen, J., Liu, X. & Nefzger, C. M. Propagation and maintenance of mouse embryonic stem cells. *Methods Mol. Biol*. **1940**, 33–45 (2019).
37. Rostovskaya, M., Stirparo, G. G. & Smith, A. Capacitation of human naïve pluripotent stem cells for multi-lineage differentiation. *Development* **146**, dev172916 (2019).
38. Boroviak, T. et al. Single cell transcriptome analysis of human, marmoset and mouse embryos reveals common and divergent features of preimplantation development. *Development* **145**, dev167833 (2018).
39. Lim, H. Y. G. et al. Keratins are asymmetrically inherited fate determinants in the mammalian embryo. *Nature* **585**, 404–409 (2020).
40. Gerri, C. et al. Initiation of a conserved trophectoderm program in human, cow and mouse embryos. *Nature* **587**, 443–447 (2020).
41. Aberkane, A. et al. Expression of adhesion and extracellular matrix genes in human blastocysts upon attachment in a 2D co-culture system. *Mol. Hum. Reprod*. **24**, 375–387 (2018).
42. Bankhead, P. et al. QuPath: Open source software for digital pathology image analysis. *Sci. Rep*. **7**, 16878 (2017).
43. Lam, A. Q. et al. Rapid and efficient differentiation of human pluripotent stem cells into intermediate mesoderm that forms tubules expressing kidney proximal tubular markers. *J. Am. Soc. Nephrol*. **25**, 1211–1225 (2014).
44. Dobin, A. et al. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics* **29**, 15–21 (2013).
45. Huang, Y., McCarthy, D. J. & Stegle, O. Vireo: Bayesian demultiplexing of pooled single-cell RNA-seq data without genotype reference. *Genome Biol*. **20**, 273 (2019).
46. Wickham, H. et al. dplyr: a grammar of data manipulation. R version 1.0.5 https://CRAN.R-project.org/package=dplyr (2021).
47. Butler, A., Hoffman, P., Smibert, P., Papalexi, E. & Satija, R. Integrating single-cell transcriptomic data across different conditions, technologies, and species. *Nat. Biotechnol*. **36**, 411–420 (2018).
48. Stuart, T. et al. Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902 (2019).
49. Wickham, H. *ggplot2: Elegant Graphics for Data Analysis* (Springer, 2016).
50. Kolde, R. & Vilo, J. GOsummaries: an R package for visual functional annotation of experimental data. *F1000Res*. **4**, 574 (2015).
51. Hafemeister, C. & Satija, R. Normalization and variance stabilization of single-cell RNA-seq data using regularized negative binomial regression. *Genome Biol*. **20**, 296 (2019).
52. Scialdone, A. et al. Computational assignment of cell-cycle stage from single-cell transcriptome data. *Methods* **85**, 54–61 (2015).
53. Linneberg-Agerholm, M. et al. Naïve human pluripotent stem cells respond to Wnt, Nodal and LIF signalling to produce expandable naïve extra-embryonic endoderm. *Development* **146**, dev180620 (2019).

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Author contributions J.M.P. and X.L. conceptualized the study and designed the experiments. X.L. and J.P.T. performed reprogramming experiments and generation of human iBlastoids, and collection and isolation of single cells for scRNA-seq and functional characterization of iBlastoids with support from Y.B.Y.S., J.C., G.S., M.M. and Y.Z. under the supervision of J.M.P. and with advice from A.T.C. D.P. helped with sequencing of iBlastoid scRNA-seq libraries under the supervision of R.L. X.L. and J.P.T. performed the immunostaining of iBlastoids with help from A.A. and J.Z. with image acquisition and analysis. J.S. performed the scRNA-seq analyses for the iBlastoids as well as the integration across the human blastocyst datasets with support from X.L., J.F.O., J.M.P. and O.J.L.R. J.F.O. and O.J.L.R. developed the interface for the interactive online tool. S.M.L. assisted with the application for ethical approval. X.L., J.P.T. and J.M.P. wrote the manuscript with input from R.L., A.T.C., O.J.L.R. and J.Z. All authors approved of and contributed to the final version of the manuscript.

Competing interests Although it is not directly related to this manuscript, O.J.L.R. and J.M.P. are co-inventors on a patent (WO/2017/106932) and are co-founders and shareholders of Mogrify Ltd., a cell therapy company. X.L., J.P.T. and J.M.P. are co-inventors on two provisional patent applications (application numbers: 2020904338 and 2020904340) filed by Monash University in relation to the generation of iBlastoids and derivation of stem cell lines from iBlastoids. The other authors declare no competing interests.

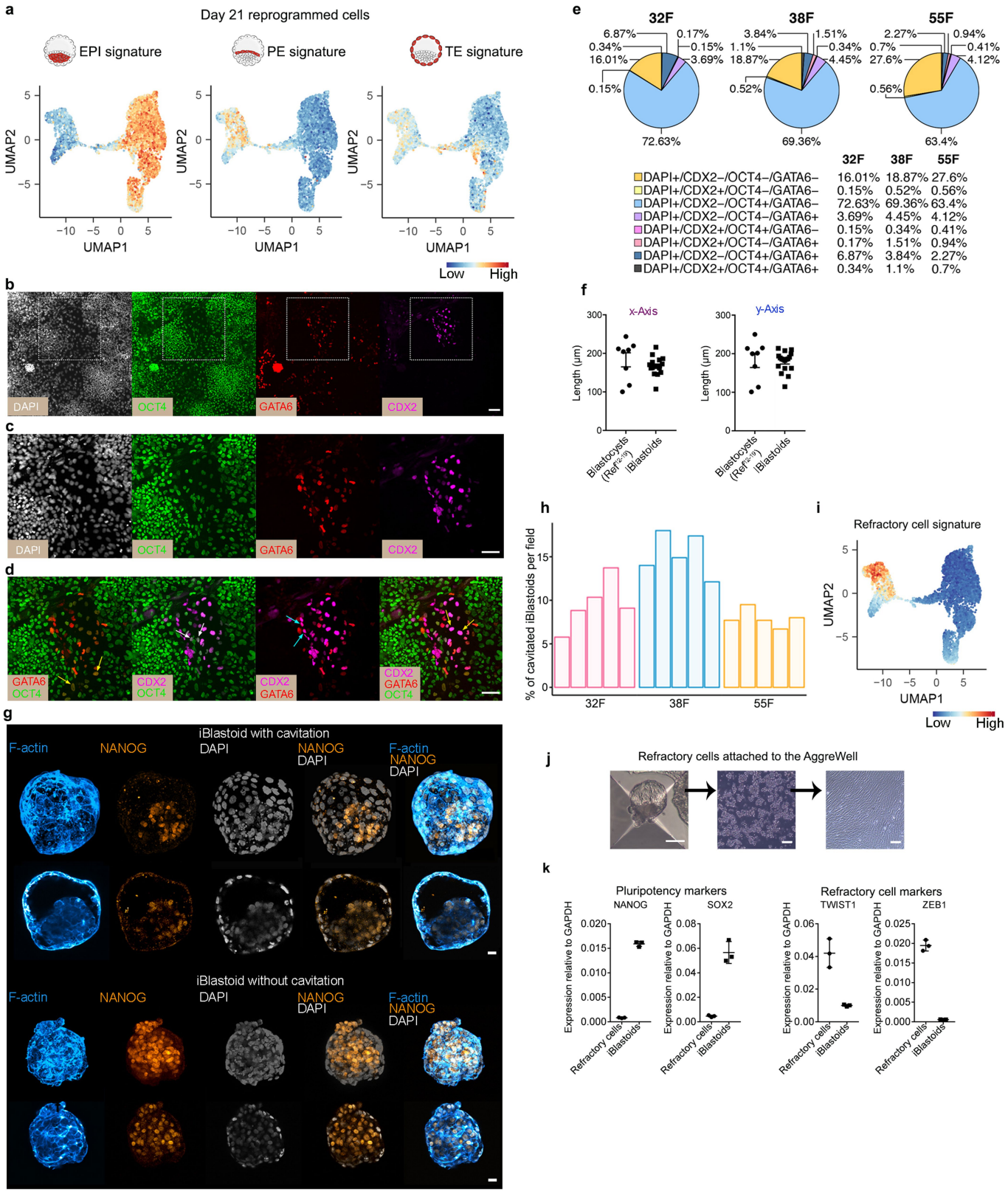
Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-021-03372-y.

Correspondence and requests for materials should be addressed to J.M.P.

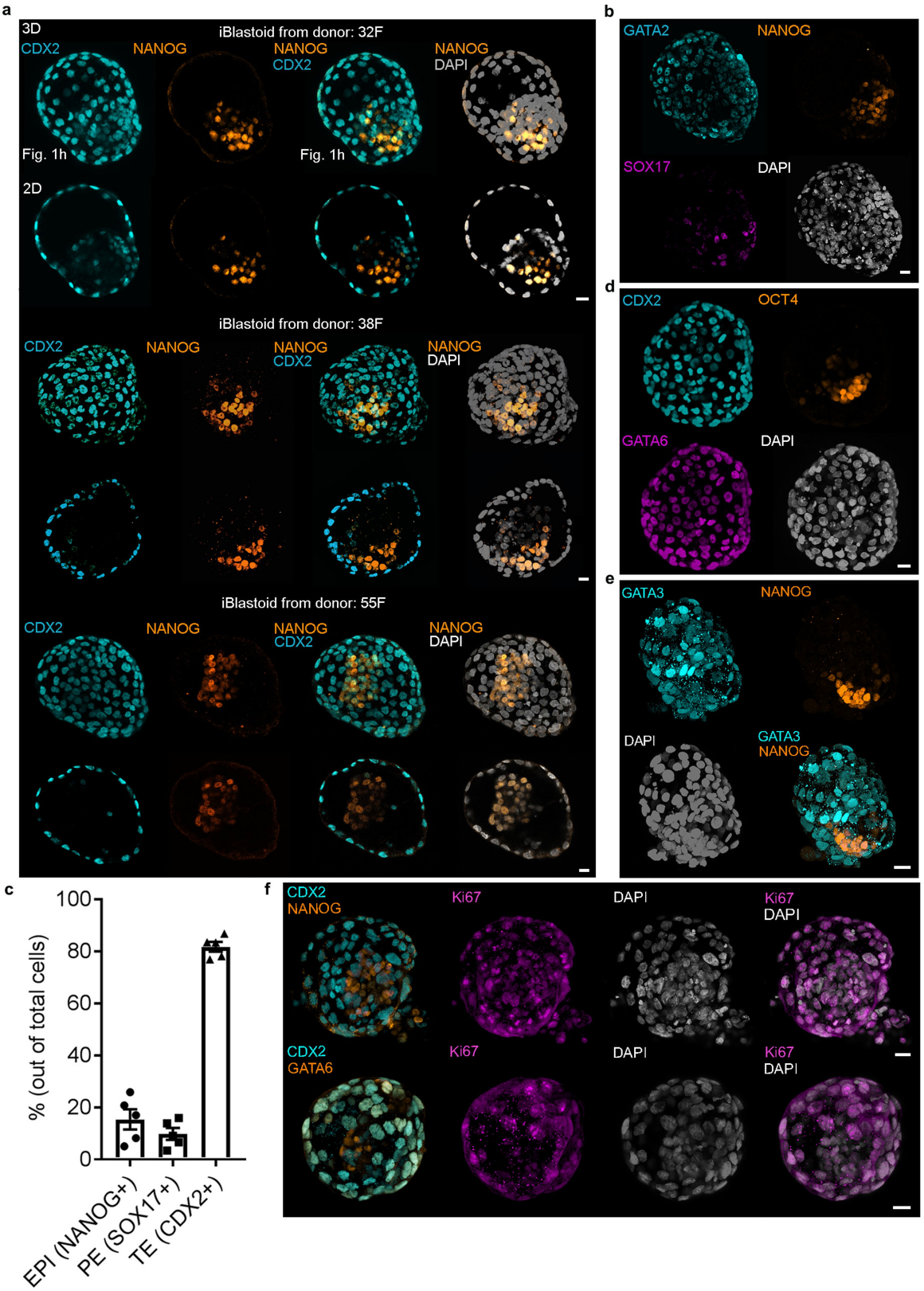
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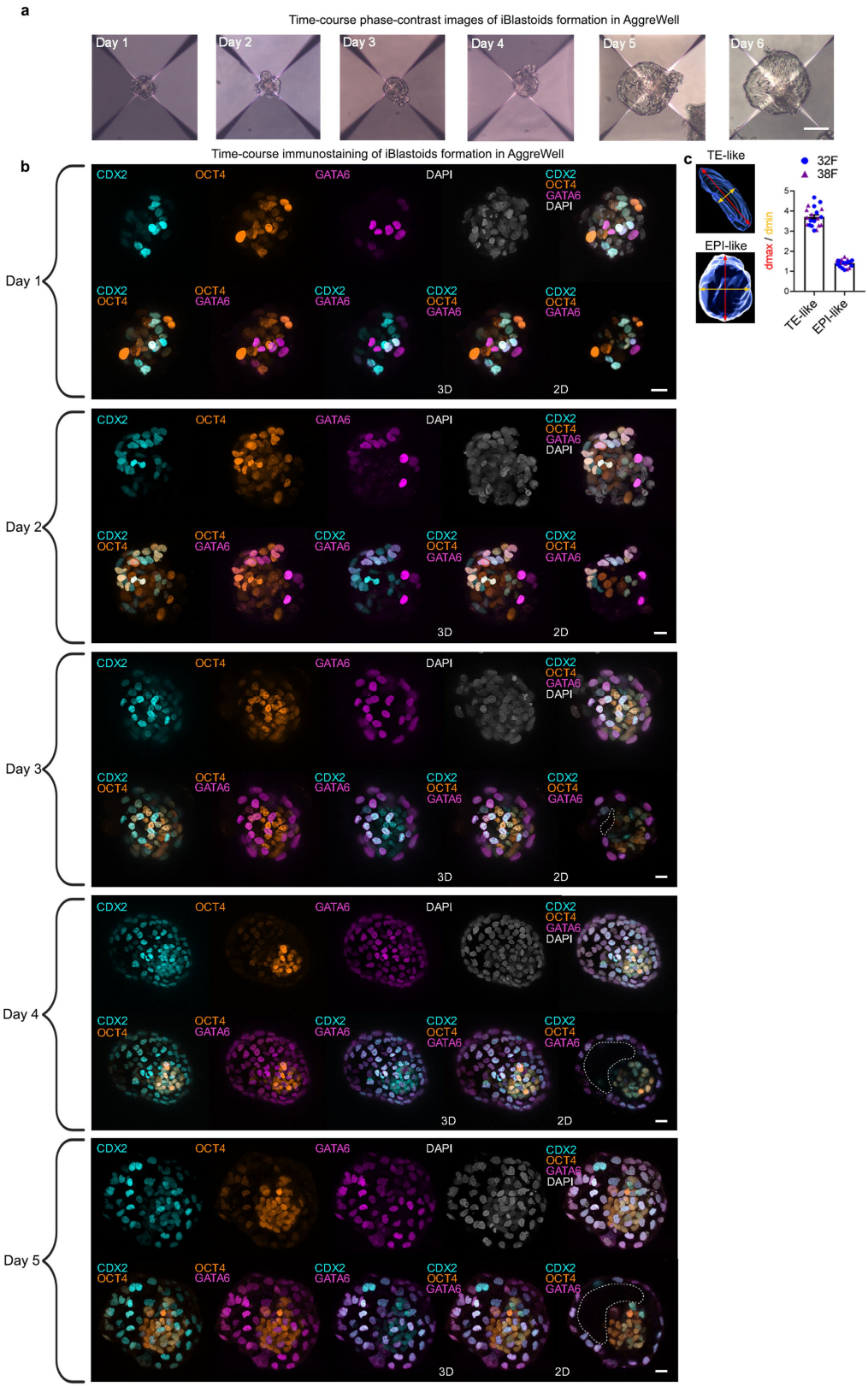


Extended Data Fig. 1 | See next page for caption.

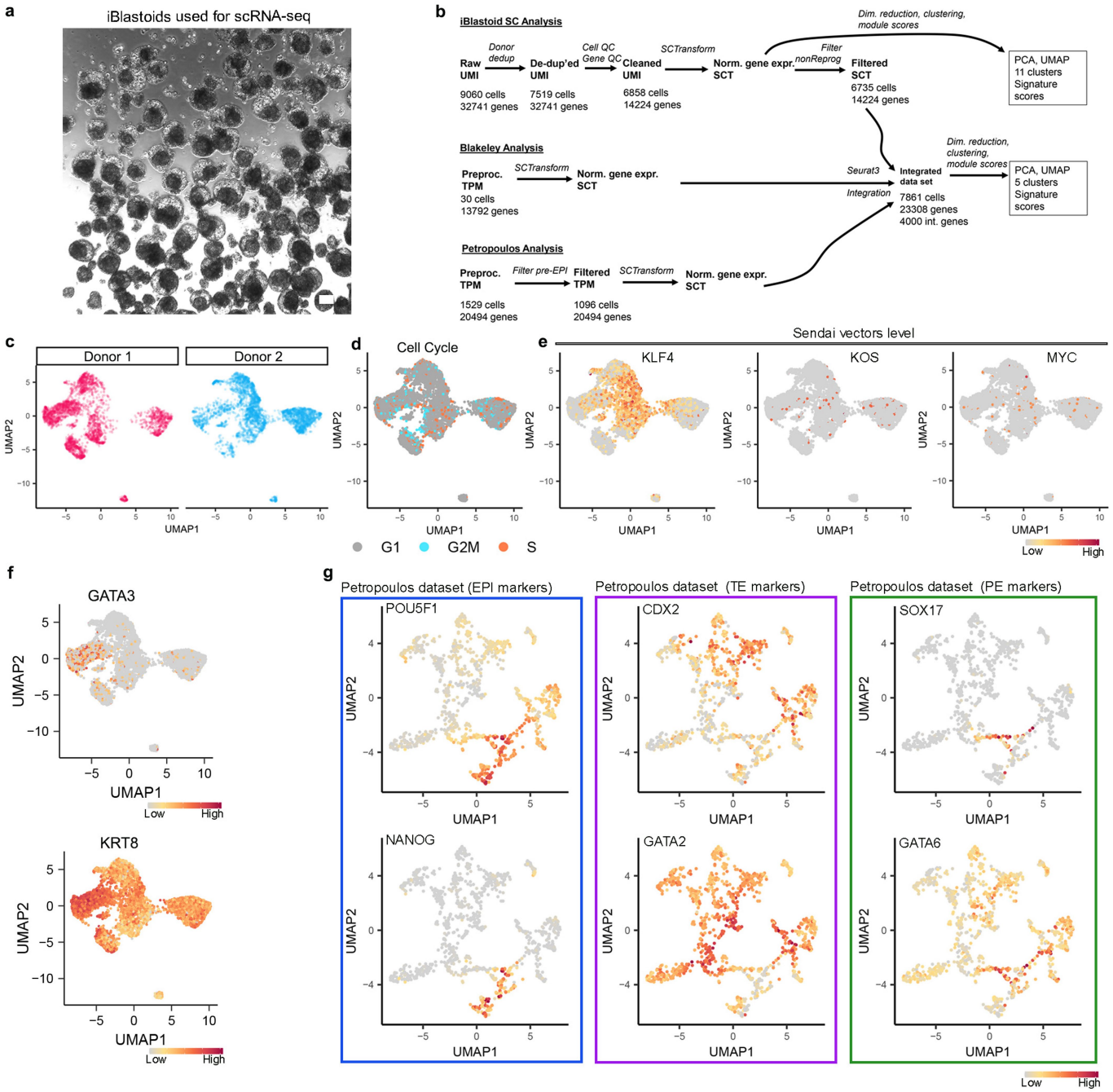
Extended Data Fig. 1 | Characterization of day 21 reprogrammed cells and morphological assessment of iBlastoids. a, scRNA-seq analysis of day 21 reprogrammed cells (6,816 cells) showing presence of EPI-like, TE-like and PE-like populations. **b**–**d**, Immunostaining of day 21 reprogrammed cells for OCT4, GATA6 and CDX2 (**b**) with expanded view of the region enclosed by dashed line (**c**) showing $OCT4^+GATA6^+CDX2^-$ cells (yellow arrows), $OCT4^+GATA6^-CDX2^+$ cells (white arrows), $OCT4^-GATA6^+CDX2^+$ cells (cyan arrows) and $OCT4^+GATA6^+CDX2^+$ cells (orange arrows) (**d**; $n = 3$). Scale bars, 100 µm. **e**, Quantification of different existing cell populations in day 21 reprogrammed cells in **b**; $n = 3$. **f**, Measurement of *x*- and *y*-axis diameters of iBlastoids compared to blastocysts[12–19]; mean ± s.e.m., $n = 18$ for iBlastoids, 8 for blastocysts[12–19]. **g**, Representative iBlastoids with and without cavitation; $n = 2$. Scale bars, 20 µm. **h**, Quantification of iBlastoids with cavity formation, $n = 628$. **i**, Per-cell expression score for refractory cell signatures on UMAP of the scRNA-seq dataset for day 21 reprogrammed cells. **j**, Refractory cells adhering to microwell edges during iBlastoid formation, which propagated in fibroblast medium with classic fibroblast morphology; $n = 2$. Scale bars, 100 µm (arrows indicate step-wise extraction and culture of refractory cells). **k**, qRT–PCR analysis of refractory cells in **j** for pluripotency markers (left) and refractory cell markers (right); mean ± s.e.m., $n = 3$. For more details on sample numbers, see 'Statistics and Reproducibility'.



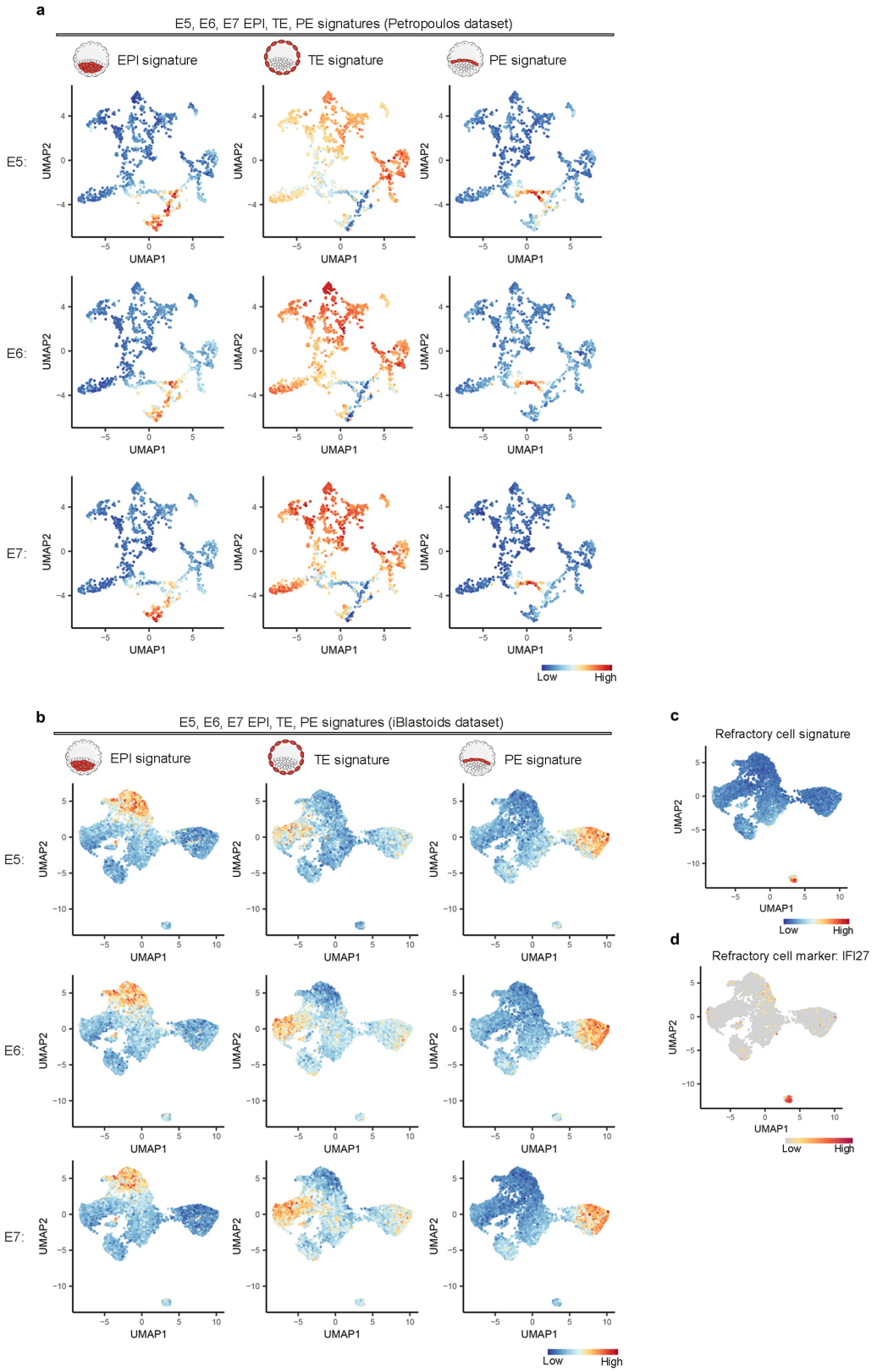
Extended Data Fig. 2 | Immunostaining analysis of iBlastoids. a, 3D and 2D representations of iBlastoids stained for CDX2 and NANOG; $n = 5$. **b**, iBlastoid stained for GATA2, NANOG, and SOX17, related to Fig. 1k; $n = 2$. **c**, Proportions of EPI-, PE- and TE-like cells out of total cells in iBlastoids; mean ± s.e.m., $n = 5$. **d**, iBlastoid stained for CDX2, OCT4, and GATA6, related to Fig. 1l; $n = 2$. **e**, iBlastoid stained for GATA3 and NANOG; $n = 2$. **f**, Assessment of proliferation marker Ki67 on iBlastoids stained for CDX2 and NANOG or CDX2 and GATA6; $n = 2$. Scale bars, 20 µm. For more details on sample numbers, see 'Statistics and Reproducibility'.



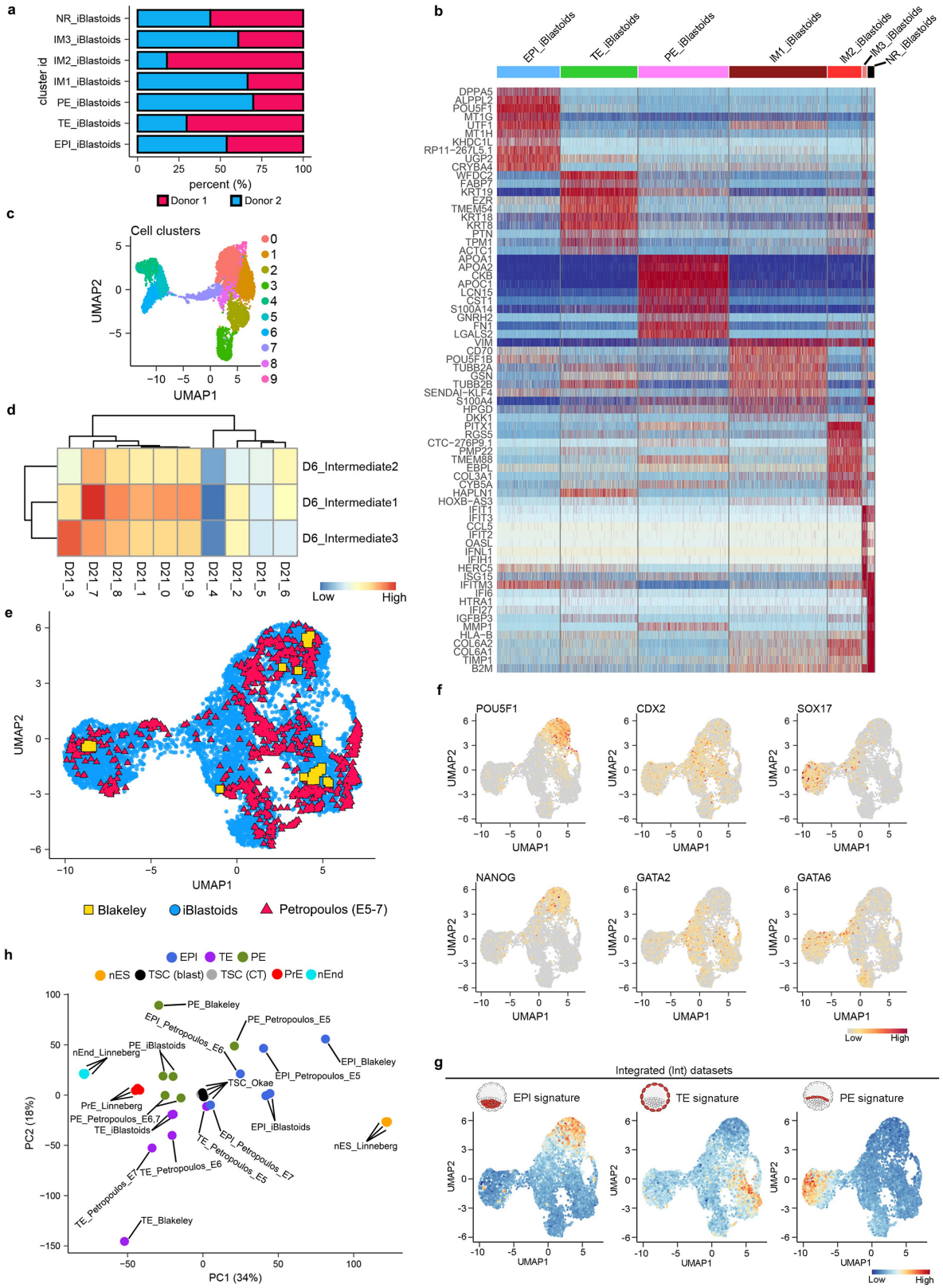
Extended Data Fig. 3 | Time-course examination of iBlastoid formation. **a**, Phase-contrast images of iBlastoid formation from day 1 to day 6; $n = 5$. Scale bar, 100 µm. **b**, Immunostaining of iBlastoids on days 1–5 for CDX2, OCT4 and GATA6; $n = 2$. Scale bars, 20 µm. **c**, Ratio of longest diameter (dmax) to shortest diameter (dmin) for TE-like and EPI-like cells in iBlastoids; mean ± s.e.m., $n = 20$. For more details on sample numbers, see 'Statistics and Reproducibility'.



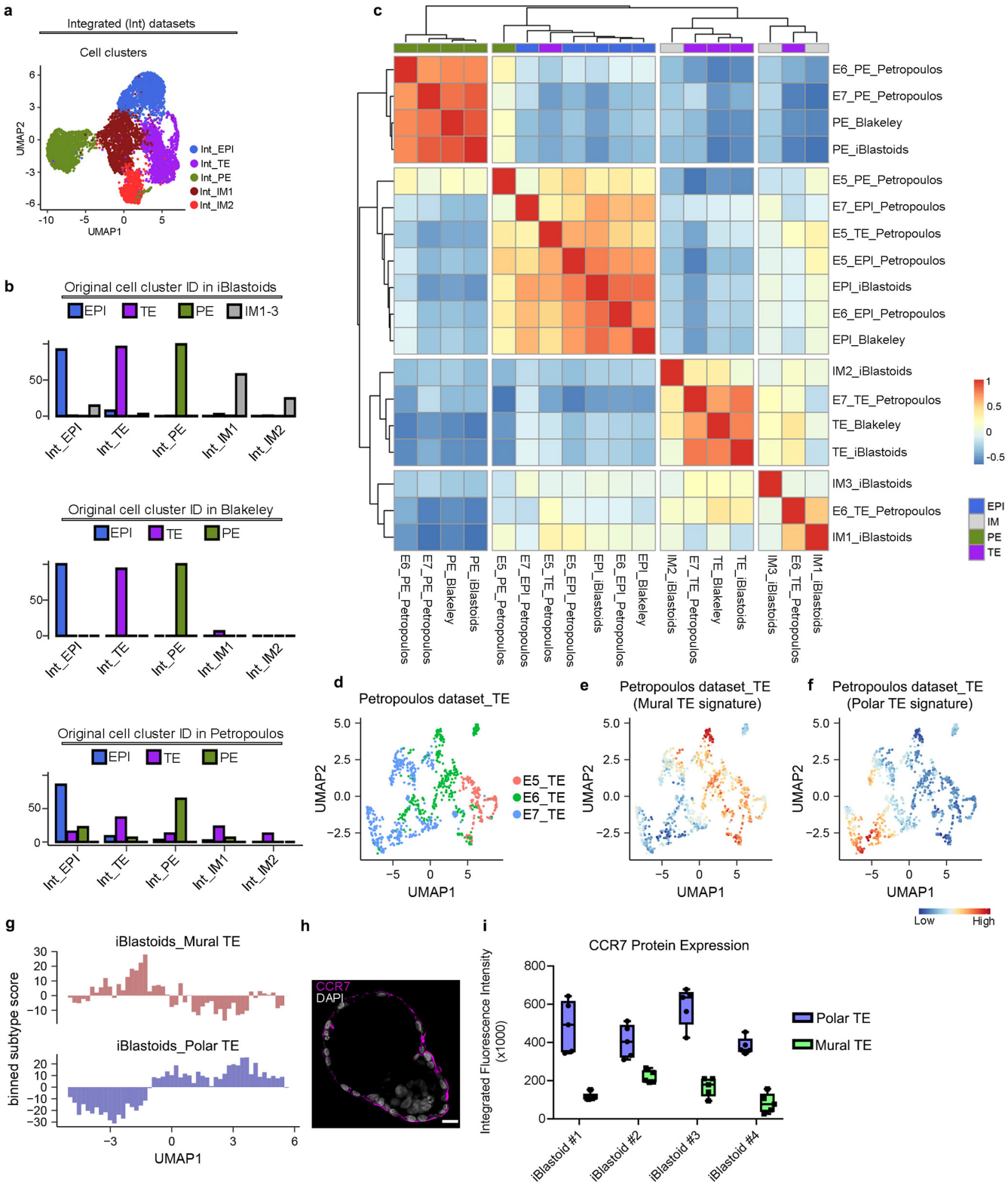
Extended Data Fig. 4 | scRNA-seq pipeline and quality controls. **a**, Phase-contrast images of the iBlastoids selected for scRNA-seq; $n = 200$. Scale bar, 100 µm. **b**, scRNA-seq analysis strategy (see Methods). **c**, UMAP representation of donor cell distribution for iBlastoids. **d**, UMAP representation of the cell cycle stages for iBlastoid scRNA-seq library. **e**, Expression of Sendai-*KLF4*, Sendai-*KOS* and Sendai-*MYC* in iBlastoids. **f**, Expression of *GATA3* and *KRT8* on UMAP of iBlastoid dataset. **g**, Expression of EPI markers (*POU5F1* and *NANOG*), TE markers (*CDX2* and *GATA2*), and PE markers (*SOX17* and *GATA6*) for Petropoulos scRNA-seq library[13]. For more details on sample numbers, see 'Statistics and Reproducibility'.



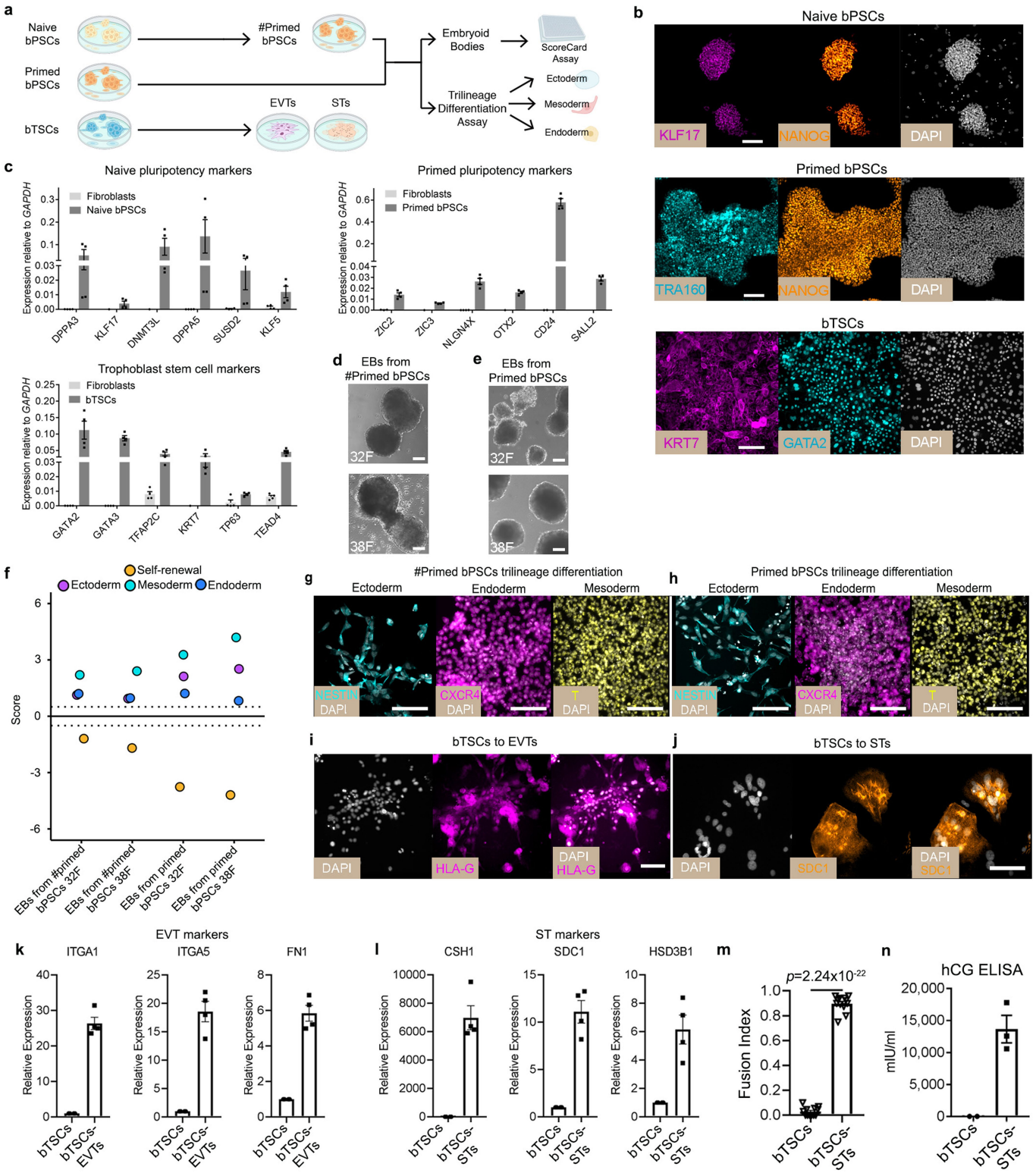
Extended Data Fig. 5 | Scoring of the defined E5–7 EPI, TE, and PE signatures for blastocysts and iBlastoids. a, Defined EPI, TE, and PE signatures for E5, E6, and E7 on Petropoulos dataset[13]. **b**, Defined EPI, TE, and PE signatures for E5, E6, and E7 on iBlastoid scRNA-seq dataset. **c**, **d**, Expression of refractory cell signature (**c**) and *IFI27* expression (**d**) on UMAP of iBlastoid dataset. For more details on sample numbers, see 'Statistics and Reproducibility'.



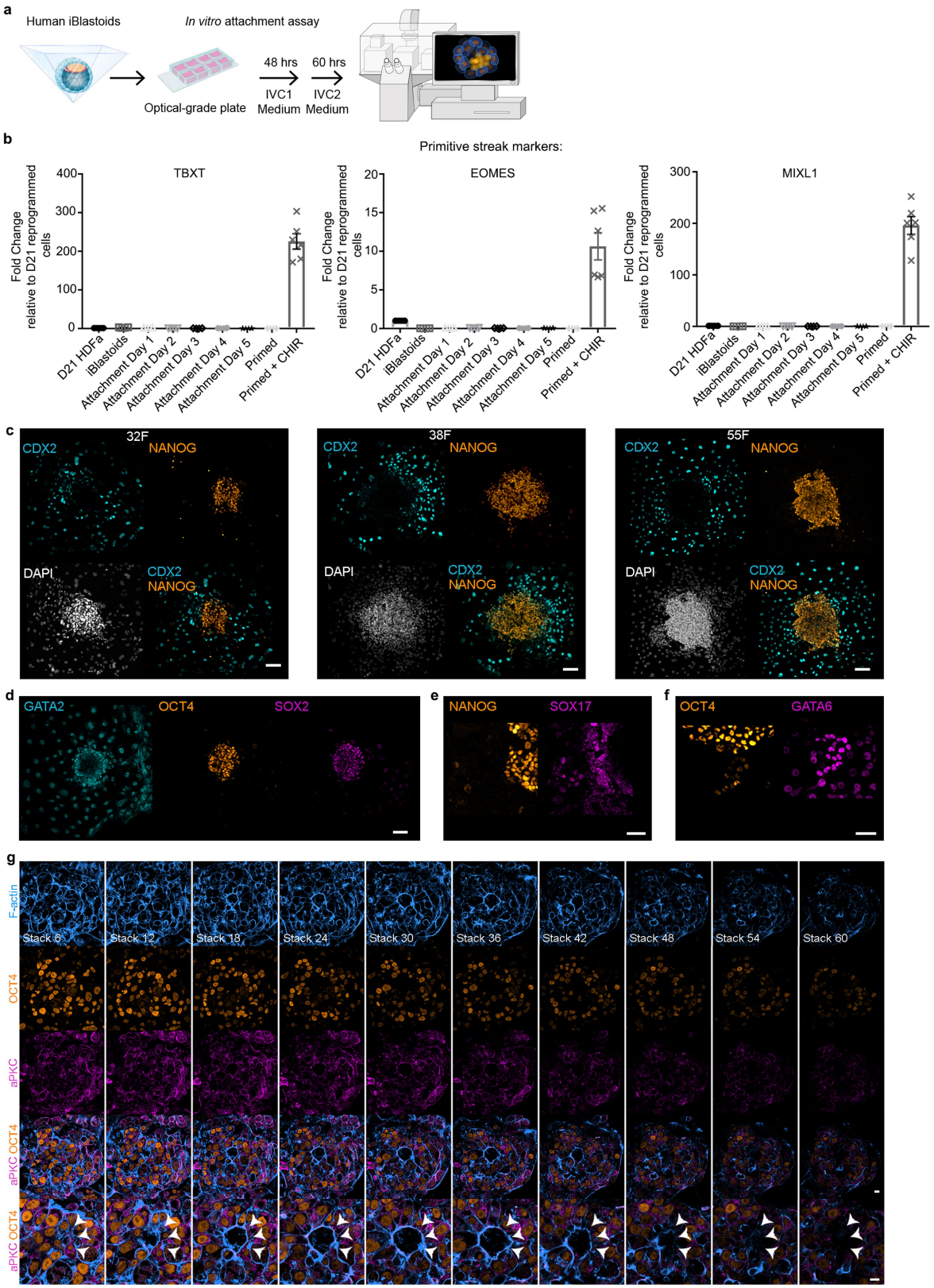
Extended Data Fig. 6 | scRNA-seq analysis of iBlastoid and blastocyst datasets. a, Proportion of cells from each donor across all assigned clusters in the iBlastoid scRNA-seq dataset. **b**, Heat map showing gene expression profiles of each assigned cluster (top 10 genes each) in the iBlastoid scRNA-seq dataset. **c**, Unsupervised clustering of day 21 reprogrammed cells in the scRNA-seq dataset. **d**, Pearson correlation analysis for clusters from day 21 reprogrammed cells with annotated IM clusters from the iBlastoid scRNA-seq dataset. **e**, UMAP projection showing cell distribution of iBlastoids and blastocysts[12,13] in the integrated dataset of iBlastoids and E5–7 blastocysts[12,13]. **f**, Expression of EPI markers (*POU5F1* and *NANOG*), TE markers (*CDX2* and *GATA2*), and PE markers (*SOX17* and *GATA6*) for the integrated dataset. **g**, Per-cell expression score of EPI, TE, and PE signatures in the integrated dataset. **h**, PCA plot of pseudo-bulk RNA-seq data from scRNA-seq dataset of iBlastoids and blastocysts[12,13] integrated with RNA-seq dataset of PS, TS and nEnd cell lines[25,53]. For more details on sample numbers, see 'Statistics and Reproducibility'.



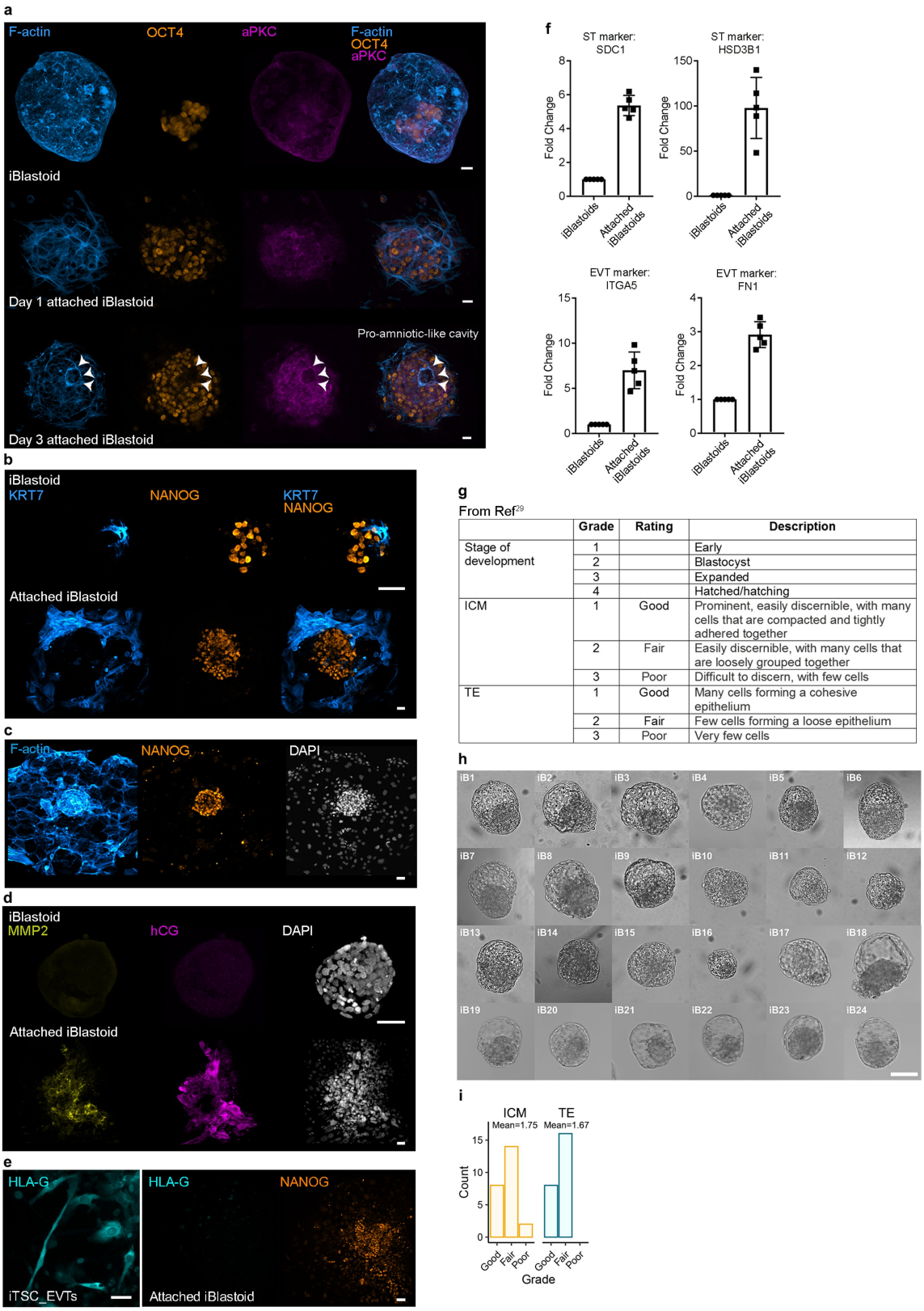
Extended Data Fig. 7 | Assessment of mural and polar TE gene expression in iBlastoid scRNA-seq dataset. a, Unsupervised clustering of the integrated dataset with assigned cluster names. **b**, Proportion of cells for iBlastoid and human blastocyst datasets[12,13] within each integrated cluster with respective original cell ID before integration analysis. **c**, Pearson correlation analysis of iBlastoid EPI, TE, and PE clusters with annotated EPI, TE, and PE clusters from blastocysts[12,13]. **d**, UMAP representation of E5–E7 TE cells of Petropoulos dataset[13]. **e**, **f**, Defined mural (**e**) and polar (**f**) TE signatures for the Petropoulos dataset[13]. **g**, Binned subtype scores for mural and polar TE signatures along UMAP component 1. **h**, Immunostaining of iBlastoid for CCR7; $n = 4$. Scale bar, 20 µm. **i**, CCR7 fluorescence intensity of polar and mural TE cells on iBlastoids; $n = 4$. Line, median; whiskers, maxima and minima; box limits, 25th to 75th percentiles. For more details on sample numbers, see 'Statistics and Reproducibility'.



Extended Data Fig. 8 | Characterization of stem cells derived from iBlastoids. a, Schematic of the experimental design for characterization of naive bPSCs, primed bPSCs and bTSCs. **b**, Immunostaining of naive bPSCs for KLF17 and NANOG, primed bPSCs for TRA160 and NANOG and bTSCs for KRT7 and GATA2, related to Fig. 3b–g; $n = 2$. **c**, qRT–PCR analysis of naive bPSCs for naive pluripotency markers, primed bPSCs for primed pluripotency markers and bTSCs for TSC markers; mean ± s.e.m., $n = 4$. **d**, **e**, EBs derived from #primed bPSCs (**d**) and primed bPSCs (**e**); $n = 2$. # indicates primed bPSCs generated from naive bPSCs. **f**, ScoreCard Assay analysis of EBs from #primed bPSCs and primed bPSCs; $n = 2$. **g**, **h**, Differentiation of #primed bPSCs (**g**) and primed bPSCs (**h**) into ectoderm, endoderm and mesoderm lineages; $n = 2$. **i**, Immunostaining of bTSC-differentiated EVTs for HLA-G; $n = 2$. **j**, Immunostaining of bTSC-differentiated STs for SDC1; $n = 2$. **k**, qRT–PCR analysis of bTSC-differentiated EVTs for EVT markers (*ITGA1*, *ITGA5*, *FN1*); mean ± s.e.m., $n = 4$. **l**, qRT–PCR analysis of bTSC-differentiated STs for ST markers (*CSH1*, *SDC1*, *HSD3B1*); mean ± s.e.m., $n = 4$. **m**, Fusion index of bTSC-differentiated STs; mean ± s.e.m., $n = 12$, two-tailed unpaired Student's *t*-test. **n**, hCG protein level detected by hCG ELISA using conditioned medium collected from bTSC-differentiated STs; mean ± s.e.m., $n = 2$–3. Scale bars, 100 µm. For more details on sample numbers, see 'Statistics and Reproducibility'.



Extended Data Fig. 9 | Assessment of epiblast development in attached iBlastoids. a, Schematic of the experimental design for iBlastoid in vitro attachment assay. **b**, Time-course qRT–PCR analysis of primitive streak markers (*TBXT*, *EOMES*, and *MIXL1*) in iBlastoids for up to day 5 in the attachment assay; mean ± s.e.m., $n = 6$. Positive control for *TBXT*, *EOMES*, and *MIXL1* was generated using a previously published mesoderm differentiation protocol[43]. **c**, Attached iBlastoids stained for CDX2 and NANOG; $n = 5$. Scale bars, 100 µm. **d**, Attached iBlastoids stained for GATA2, OCT4, and SOX2; $n = 2$. Scale bars, 100 µm. **e**, Expanded view of attached iBlastoids for NANOG and SOX17 staining, related to Fig. 4b; $n = 2$. **f**, Expanded view of attached iBlastoids for OCT4 and GATA6 staining, related to Fig. 4c; $n = 2$. Scale bars for **e**, **f**, 50 µm. **g**, *Z*-section series of attached iBlastoids stained with F-actin, OCT4 and aPKC; bottom, expanded view of pro-amniotic cavity. $n = 2$. Scale bar, 10 µm. Arrows mark appearance of the pro-amniotic-like cavity. For more details on sample numbers, see 'Statistics and Reproducibility'.



	Grade	Rating	Description
Stage of development	1		Early
	2		Blastocyst
	3		Expanded
	4		Hatched/hatching
ICM	1	Good	Prominent, easily discernible, with many cells that are compacted and tightly adhered together
	2	Fair	Easily discernible, with many cells that are loosely grouped together
	3	Poor	Difficult to discern, with few cells
TE	1	Good	Many cells forming a cohesive epithelium
	2	Fair	Few cells forming a loose epithelium
	3	Poor	Very few cells

Extended Data Fig. 10 | Characterization of attached iBlastoids. **a**, Immunostaining of iBlastoids, day 1 attached iBlastoids and day 3 attached iBlastoids for F-actin, OCT4 and aPKC; $n = 2$. Scale bars, 20 µm. Arrows mark appearance of the pro-amniotic-like cavity. **b**, Immunostaining of iBlastoids and attached iBlastoids for KRT7 and NANOG, related to Fig. 4e; $n = 4$. **c**, Attached iBlastoids stained for F-actin and NANOG, related to Fig. 4f; $n = 2$. **d**, Immunostaining of iBlastoids and attached iBlastoids for MMP2 and hCG, related to Fig. 4g; $n = 2$. **e**, Attached iBlastoids stained for HLA-G and NANOG with induced trophoblast stem cell-differentiated EVTs as positive control for HLA-G immunostaining (left). Scale bars for **b**–**e**, 50 µm. **f**, qRT–PCR analysis of ST markers (*SDC1*, *HSD3B1*; top) and EVT markers (*ITGA5*, *FN1*; bottom) in attached iBlastoids; mean ± s.e.m., $n = 5$. **g**, Assessment criteria[29] used for scoring of iBlastoids. **h**, Phase-contrast images of iBlastoids included for scoring assessment; $n = 24$. Scale bar, 100 µm. **i**, Average grades of ICM and TE for iBlastoids in **h** according to **g**; $n = 24$. For more details on sample numbers, see 'Statistics and Reproducibility'.

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Corresponding author(s): Jose M. Polo

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☒	☐	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐	Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Motic Image Plus 2.0, LightCycler 480 software (Roche, version 1.5.1.62), BD FACS TM software, version 1.2.0.142 for influx sorters, Leica application suite X version 3.5.6.21594 for SP8, Leica application suite X version 3.7.1.21655 for DMi8, ZEISS Efficient Navigation version 6.0.0.485 for ZEISS LSM 780.
Data analysis	Data was analyzed using: GraphPad Prism (v7), ImageJ (v1.8.0_112), Imaris x64 version 9.5.0, QuPath (version 0.2.3), Adobe Photoshop (version 21.2.3), Cellranger (v2.1.0, v3.0.2, v3.1.0), base R(3.6.3), Seurat 3.1.5, Cellranger 3.1.0, STAR aligner (v2.5.1b), limma 3.42.2, tidyverse 1.3.0, RColorBrewer 1.1-2, ggplot2 3.3.1, egg 0.4.5, shiny 1.2.0, cellSNP 0.3.0, vireoSNP 0.3.2, pheatmap (v1.0.12). All data were analysed with standard programs and packages as detailed above. Scripts can be found at http://github.com/jibsch/iBlastoids_scripts.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors/reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Research guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A list of figures that have associated raw data
- A description of any restrictions on data availability

All processed data sets are available at (http://blastoid.ddnetbio.com/). Raw and processed next generation sequencing datasets of iBlastoids were deposited at the Gene Expression Omnibus (GEO) repository with the following accession numbers: GSE156596. Published human embryo datasets used in this study were obtained from Blakeley et al and Petropoulos et al under accession numbers GSE66507 and E-MTAB-3929 (ArrayExpress) respectively. For Extended Data Fig. 6h, datasets

were abstained from Okae et al Table S1, Linneberg-Agerholm et al: GSE138012 for the RNA-seq integration. All source data related to this study are available in the source data table related to each figure, which are provided within the online content of this paper. There are no restrictions on data availability.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We did not involve statistical methods to pre-determine the sample size, this was determined based on previous experience and other similar studies. We generated iBlastoids using three different donor fibroblasts in multiple runs of reprogramming experiments (at least 4 independent replicates). Furthermore, we performed transcriptional profiling of iBlastoids using scRNA-seq libraries (from 2 donors). Similar results/findings were observed. These suggest that our sample size is sufficient.
Data exclusions	No data were excluded.
Replication	Each experiment was reproduced at least in 4 biological replicates (from two donors) if not otherwise stated. Please refer to figure legends and methods for details. All replications were successful.
Randomization	The iBlastoids generated were pre-selected based on morphological assessment, however for all downstream analyses the pre-selected iBlastoids were chosen at random. The experiments were not randomized.
Blinding	The investigators were not blinded during data collection and analysis, as neither human/animal studies or specific grouping were involved in this manuscript.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☐	☒ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology
☒	☐ Animals and other organisms
☒	☐ Human research participants
☒	☐ Clinical data

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used

Details of all antibodies used in this study were provided in Supplementary Table 9 with catalog number, lot number and commercial sources supplied.

For Immunostaining:
Antibody, Company, Catalogue Number, Lot Number, Dilution:
Mouse anti-GATA2 IgG2a Sigma-Aldrich Cat# WH0002624M1 Lot G2151-2D11 1:100
Rabbit anti-CK7 (KRT7) IgG abcam Cat# ab181598 Lot GR3214132-7 1:100
Mouse anti-CDX2 IgG1 abcam Cat# ab157524 Lot GR3301072-2 1:50
Rabbit anti-CDX2 IgG Cell Signaling Technology Cat# 12306 Lot 2 1:100
Rabbit anti-NANOG polyclonal abcam Cat# ab21624 Lot GR3260262-26 1:100
Mouse anti-NANOG IgG1 Invitrogen Cat# 14-5768-82 Lot 2003073 1:100
Mouse anti-OCT3/4 IgG2b Santa Cruz Biotechnology Cat# sc-5279 Lot I3015 1:50
Mouse anti-SOX17 IgG1 abcam Cat# ab84990 Lot GR3277101-3 1:100
Rat anti-SOX2 IgG2a Invitrogen Cat# 14-9811-82 Lot 2124609 1:200
Goat anti-GATA6 polyclonal R&D Systems Cat# AF1700 Lot KWT0418111 1:200
Rabbit anti-MMP2 IgG Cell Signaling Technology Cat# 40994 Lot 3 1:800
Mouse anti-hCG IgG1 abcam Cat# ab9582 Lot GR3285769-1 1:100
Alexa Fluor 488 Phalloidin ThermoFisher Cat# A-12379 Lot 2069633 1:200

Rabbit anti-Laminin IgG abcam Cat# ab11575 Lot GR3268366-6 1:100
Mouse anti-PKC ζ IgG2a Santa Cruz Biotechnology Cat# sc-17781 Lot F2520 1:100
Rabbit anti-GATA3 IgG abcam Cat# ab199428 Lot GR3266930-1 1:100
Rat anti-KRT8 IgG2a Merck Millipore Cat# MABT329 Lot 3429042 1:100
Rabbit anti-CCR7 IgG abcam Cat# ab32527 Lot GR259329-31 1:100
Mouse anti-Nestin IgG1 Merck Millipore Cat# MAB5326 Lot 2393925 1:100
Mouse anti-CXCR4 IgG2a-PE Miltenyi Biotec Cat# 130-117-690 Lot 5200108213 1:100
Rabbit anti-Brachyury IgG Cell Signaling Technology Cat# 81694 Lot 1 1:200
Mouse anti-HLA-G IgG1 [MEM-G/1] abcam Cat# ab7759 Lot GR3262011-5 1:50
Rabbit anti-SDC1 IgG Cell Signaling Technology Cat# 12922 Lot 1 1:400
Rabbit anti-Ki67 IgG abcam Cat# ab15580 Lot GR3198185-1 1:100

Goat anti-rat IgG AF555 secondary ThermoFisher Cat# A-21434 Lot 1726587 1:500
Goat anti-rabbit IgG AF555 secondary ThermoFisher Cat# A-21428 Lot 1786491 1:500
Goat anti-mouse IgG2b AF647 secondary ThermoFisher Cat# A-21242 Lot 1968702 1:500
Goat anti-mouse IgG-AF488 secondary ThermoFisher Cat# A-11029 Lot 2066710 1:500
Goat anti-mouse IgG1-AF488 secondary ThermoFisher Cat# A-21121 Lot 1964382 1:500
Goat anti-mouse IgG2a-AF488 secondary ThermoFisher Cat# A-21131 Lot 2136788 1:500
Goat anti-mouse IgG2a-AF555 secondary ThermoFisher Cat# A-21137 Lot 1899521 1:500
Goat anti-mouse IgG2a-AF647 secondary ThermoFisher Cat# A-21241 Lot 2056280 1:500
Donkey anti-mouse IgG-488 secondary ThermoFisher Cat# A-21202 Lot 1696430 1:500
Donkey anti-goat IgG-555 secondary ThermoFisher Cat# A-21432 Lot 872636 1:500
Donkey anti-rabbit IgG-647 secondary ThermoFisher Cat# A-31573 Lot 1903516 1:500

Validation

Antibodies obtained from the commercial source were validated by the suppliers, detailed validation analysis relevant literatures are provided on the company website for the products used in this study. Some antibodies were validated in a previously published study as indicated in methods or relevant literature was cited.

1. GATA2 (WH0002624M1) https://www.sigmaaldrich.com/catalog/product/sigma/wh0002624m1
2. KRT7 (ab181598) https://www.abcam.com/cytokeratin-7-antibody-epr17078-cytoskeleton-marker-ab181598.html
3. CDX2 (ab157524) https://www.abcam.com/cdx2-antibody-cdx2-88-ab157524.html
4. CDX2 (12306) https://www.cellsignal.com/products/primary-antibodies/cdx2-d11d10-rabbit-mab/12306
5. NANOG (ab21624) https://www.abcam.com/nanog-antibody-ab21624.html
6. NANOG (14-5768-82) https://www.thermofisher.com/antibody/product/Nanog-Antibody-clone-hNanog-2-Monoclonal/14-5768-82
7. OCT-3/4 (sc-5279) https://www.scbt.com/p/oct-3-4-antibody-c-10
8. SOX17 (ab84990) https://www.abcam.com/sox17-antibody-oti3b10-ab84990.html
9. SOX2 (14-9811-82) https://www.thermofisher.com/antibody/product/SOX2-Antibody-clone-Btjce-Monoclonal/14-9811-82
10. GATA6 (AF1700) https://www.rndsystems.com/products/human-gata-6-antibody_af1700
11. MMP2 (40994) https://www.cellsignal.com/products/primary-antibodies/mmp-2-d4m2n-rabbit-mab/40994
12. hCG (ab9582) https://www.abcam.com/hcg-beta-antibody-5h4-e2-ab9582.html
13. Alexa Fluor 488 Phalloidin (A-12379) https://www.thermofisher.com/order/catalog/product/A12379#/A12379
14. Laminin (ab11575) https://www.abcam.com/laminin-antibody-ab11575.html
15. PKC ζ (sc-17781) https://www.scbt.com/p/pkc-zeta-antibody-h-1
16. GATA3 (ab199428) https://www.abcam.com/gata3-antibody-epr16651-chip-grade-ab199428.html
17. KRT8 (MABT329) https://www.merckmillipore.com/AU/en/product/Anti-Cytokeratin-8-Antibody-clone-TROMA-1,MM_NF-MABT329
18. CCR7 (ab32527) https://www.abcam.com/ccr7-antibody-y59-ab32527.html
19. Nestin (MAB5326) https://www.sigmaaldrich.com/catalog/product/mm/mab5326?
20. CXCR4-PE (130-117-690) https://www.miltenyibiotec.com/AU-en/products/cd184-cxcr4-antibody-anti-human-12g5.html
21. Brachyury (81694) https://www.cellsignal.com/products/primary-antibodies/brachyury-d2z3j-rabbit-mab/81694
22. HLA-G (ab7759) https://www.abcam.com/hla-g-antibody-mem-g1-ab7759.html
23. SDC1 (12922) https://www.cellsignal.com/products/primary-antibodies/syndecan-1-d4y7h-rabbit-mab/12922
24. Ki67 (ab15580) https://www.abcam.com/ki67-antibody-ab15580.html
25. Goat anti-rat IgG AF555 (A-21434) https://www.thermofisher.com/antibody/product/Goat-anti-Rat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21434
26. Goat anti-rabbit IgG AF555 (A-21428) https://www.thermofisher.com/antibody/product/Goat-anti-Rabbit-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21428
27. Goat anti-mouse IgG2b AF647 (A-21242) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG2b-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21242
28. Goat anti-mouse IgG-AF488 (A-11029) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-11029
29. Goat anti-mouse IgG1-AF488 (A-21121) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG1-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21121
30. Goat anti-mouse IgG2a-AF488 (A-21131) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG2a-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21131
31. Goat anti-mouse IgG2a-AF555 (A-21137) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG2a-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21137
32. Goat anti-mouse IgG2a-AF647 (A-21241) https://www.thermofisher.com/antibody/product/Goat-anti-Mouse-IgG2a-Cross-

Adsorbed-Secondary-Antibody-Polyclonal/A-21241
33. Donkey anti-mouse IgG-488 (A-21202) https://www.thermofisher.com/antibody/product/Donkey-anti-Mouse-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21202
34. Donkey anti-goat IgG-555 (A-21432) https://www.thermofisher.com/antibody/product/Donkey-anti-Goat-IgG-H-L-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-21432
35. Donkey anti-rabbit IgG-647 (A-31573) https://www.thermofisher.com/antibody/product/Donkey-anti-Rabbit-IgG-H-L-Highly-Cross-Adsorbed-Secondary-Antibody-Polyclonal/A-31573

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	Human fibroblasts sourced from ThermoFisher (Catalogue number, C-013-5C and lot#1029000 for 38F and lot#1569390 for 32F) for reprogramming experiments.
Authentication	Human dermal fibroblasts were derived from tissues of donor (age and sex identified) and authenticated by ThermoFisher (via growth performance assay, testing for Hepatitis B virus, Hepatitis C virus, HIV-1 virus, mycoplasma as well as sterility testing of bacteria and fungus), as stated in the certificate of analysis. These dermal fibroblasts have also been authenticated in-house by immunostaining, qPCR, RNA-seq.
Mycoplasma contamination	Fibroblasts lines were tested by ThermoFisher. Furthermore, cell lines were regularly tested and were mycoplasma negative.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in this study.

Flow Cytometry

Plots

Confirm that:

☒ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☒ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☒ All plots are contour plots with outliers or pseudocolor plots.

☒ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	iBlastoids were dissociated with TrypLE express (ThermoFisher), and DPBS (ThermoFisher) supplemented with 2% FBS (ThermoFisher) and 10µM Y-27632 (Selleckchem) was used for final resuspension of the samples. Dissociated cells were pelleted at 400× g for 5 mins and then resuspended in a final volume of 500 µl with propidium iodide (PI) (Sigma) added to a concentration of 2µg/ml. Cell sorting was carried out with a 100 µm nozzle on an Influx instrument (BD Biosciences).
Instrument	BD Influx cell sorters (BD).
Software	Collection: FACS TM software suit (version 1.2.0.142) (BD) for influx sorters.
Cell population abundance	Abundance of distinct cell populations of interest was determined using appropriate negative controls and puritiy of sorted populations as determined by post sort reanalysis.
Gating strategy	Standard gating setting commonly utilized at flowcore facility of Monash University were used. Cell debris was excluded using a SSC vs FSC-H gate; aggregates were excluded then via a FSC-A vs FSC-H approach; dead cells were defined as PI high/positve and gated out; Live cell populations sorted were confirmed by post sort reanalysis. A figure exemplifying the FACS gating strategy is available at the Supplementary Figure 1 of the manuscript.

☒ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.